Hyperspectral Anomaly Detection by Data Sphering and Sparsity Density Peaks

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***Abstract*— Many approaches have been developed for hyperspectral anomaly detection (AD). Of particular interest are low rank and sparse representation (LRaSR) model-based methods which decompose a data space into low rank and sparse spaces. This article develops a rather different approach, which assumes that background (BKG) and anomalies can be, respectively, characterized by data statistics of the first two orders (2OS) and high orders (HOS) greater than 2. As a result, data sphering (DS) can remove BKG, while the fast independent component analysis (FastICA) can generate independent components (ICs) to form a sparse space. However, since non-Gaussian noises cannot be separated by FastICA, directly extracting anomalies from the ICs-formed sparse space may not be effective. To address this issue, a new concept of sparsity density peak (SDP) is proposed to represent the data samples in the sparse space as a probability density function from which a set of data samples with peaks can be extracted to form an anomaly space. Three versions of SDP, fixed spectral SDP (FS-SDP), fixed band SDP (FB-SDP), and spectral–spatial–sparsity peak (SS-SDP) are derived and used where the number of peaks to be extracted is determined by virtual dimensionality (VD) and a minimax-singular value decomposition (MX-SVD) algorithm. The experimental results demonstrate that DS coupled with SDP performs better than currently being used LRaSR model-based methods in AD.**

***Index Terms*— Anomaly detection (AD), data sphering (DS), independent component analysis (ICA), minimax-singular value decomposition (MX-SVD), sparsity cardinality (SC), sparsity density peak (SDP), virtual dimensionality (VD).**

List of Acronyms

2OS	First two orders of data statistics.
ATGP	Automatic target generation process.
AUC	Area under an ROC curve.
BKG	Background.
BS	Background suppression.
CEM	Constrained energy minimization.
CEM-AD	CEM anomaly detector.
CDA	Component decomposition analysis.
CRD	Collaborative representation detector.
DS	Data sphering.
DS-GoDec-SDP-AD	Anomaly detector operated in the GoDec using SDP.
DS-IC-SDP-AD	Anomaly detector operated in the data sphered IC space using SDP.
DS-SC/SDP-AD	Anomaly detector operated in the data sphered space using SC/SDP.
FastICA	Fast independent component analysis.
GoDec	Go decomposition.
IC	Independent component.
ICA	Independent component analysis.
ICSC	Independent component sparsity cardinality.
j-FS-SDP	j-fixed spectral SDP.
j-FB-SDP	j-fixed band SDP.
LRaSMD	Low rank and sparse matrix decomposition.
LRR	Low rank representation.
LSDM-MoG	Low-rank and sparse decomposition with mixture of Gaussians.
MX-SVD	Minimax singular value decomposition.
OSP	Orthogonal subspace projection.
RPCA	Robust principal component analysis.
PTA	Prior-based tensor approximation.
RX-AD	Reed-Xiaoli anomaly detector.
ROC	Receiver operating characteristic.
SC	Sparsity cardinality.
SDP	Sparsity density peak.
SS	Spectral–spatial.
SS-SDP	Spectral–spatial SDP.
VD	Virtual dimensionality.

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I. Introduction

ANOMALY detection (AD) has received considerable interests in hyperspectral imaging lately because very high spectral resolution provided by hyperspectral imagery is capable of revealing many unknown subtle material substances that cannot be identified by prior knowledge or visual inspection [1]–[3]. Consequently, detecting anomalies without prior knowledge presents a great challenge and is much more difficult than detecting known targets. Several major approaches have been investigated in the literature and can be described as follows.

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A. Categorization of AD Approaches

1) Classical Likelihood Ratio Test (LRT)-Based AD: Traditional ADs were derived from LRT using a binary hypothesis testing formulation [4] where Reed and Xiaoli extended LRT to a generalized LRT (GLRT) in [5], referred to as RX-AD. Since then, many ADs have been developed as its variants [6]–[9]. Specifically, Chang and Chiang [10] replaced the global sample auto-covariance matrix $\mathbf{K}$ with the global sample auto-correlation matrix $\mathbf{R}$ to derive R-AD which was further shown to be a special case of CEM-AD by Chang [11] where details of theoretical foundations of RX-AD and R-AD from a hyperspectral imaging perspective were derived. To further capture nonlinear spectral information, kernel-based AD (KRX-AD) was also investigated, such as support vector data description (SVDD) by Banerjee *et al.* [12], and KRX-AD by Kwon and Nasrabadi in [13].

2) Window-Based AD: Since AD is to find data samples whose spectral properties distinct from the surroundings of their neighborhoods, using the global sample covariance/correlation matrix $\mathbf{K/R}$ is generally not effective. To resolve this dilemma, many ADs have been developed to use local windows to capture local spectral characteristics such as local RX-AD by Matteoli *et al.* in [14], weighted-RX-AD by Guo *et al.* [15], a sliding window by Chang *et al.* [16], and a dual window-based eigen separation transform (DWEST) was developed by Kwon *et al.* [17], followed by collaborative representation-based detector (CRD) by Li and Du [18] which also used dual windows, and multiple-window AD (MWAD) by Liu and Chang [19] by making use of multiple windows to perform AD adaptively, guided filtering-based AD by Xie *et al.* [20] and SS feature extraction-based AD by Lei *et al.* [21].

3) BKG Separation, Estimation, and Reconstruction-Based AD: Due to a lack of prior knowledge of anomalies, BKG has a significant impact on AD. Although the abovementioned window-based AD can also capture the local BKG for each data sample, it may not be effective to characterize the global BKG, which is the key element in designing hyperspectral AD algorithms. Accordingly, in recent years, many approaches have been proposed along this line by either estimating or reconstructing BKG so that anomalies can be extracted after BKG being suppressed or removed, such as a BKG-anomaly component projection and separation optimized filter (BASO) by Chang *et al.* [22] and generative adversarial network (GAN) by Zhong *et al.* [23]. To estimate BKG, Du *et al.* [24] and Zhao *et al.* [25] proposed approaches to calculate sparsity scores for data samples, while Qu *et al.* [26] and Li *et al.* [27] estimated BKG using a mixture of Gaussians (LSDM-MoG). Most recently, deep-learning networks have also been investigated for BKG reconstruction in [28] and are also autoencoder (AE) by Borghesi *et al.* [29] and Wang *et al.* [30], GAN by Zhong *et al.* [23] and Jiang *et al.* [31], and adversarial autoencoder (AAE) by Xie *et al.* [32] and [33].

4) Low Rank and Sparse Representation (LRaSR)-Based AD: Until recently, a more interesting approach using LRaSR has attracted much attention. One earlier approach was RPCA developed by Candes *et al.* [34] and Zhou *et al.* [35] for blind source separation technique which decomposed the data matrix $\mathbf{X}$ into $\mathbf{X} = \mathbf{L} + \mathbf{S}$ in a matrix decomposition. Analogous to RPCA, Vaswani *et al.* [36] and Vaswani *et al.* [37] developed an approach, called robust subspace learning (RSL), which treated an outlier as an additive sparse corruption via sparse + low-rank (S + LR) matrix decomposition with more details available at website https://github.com/andrewssobral/lrslibrary. Using this interpretation, RPCA can be considered as a variant of RSL via S + LR formulation [38], [39]. In addition, Xu *et al.* [40] and Su *et al.* [41] proposed a LRR-based approach which decomposed a data space into a low-rank space considered to be BKG and a sparse space for anomalies by regularizing sparse representation parameters. Different from RPCA and LRR, Li *et al.* [42] took advantage of tensor Tucker decomposition to develop a PTA which decomposed a hyperspectral imagery into a BKG tensor and an anomaly tensor corresponding to BKG and anomaly spaces, respectively.

5) Low Rank and Sparse Matrix Decomposition (LRaSMD)-Based AD: One major problem with LRaSR-based AD is that it does not separate noise from sparse space. To address this issue, Zhou and Tao [43] derived a more attractive LRaSMD model which expressed a data matrix $\mathbf{X}$ as a linear decomposition of a low rank matrix, $\mathbf{L}$ and a sparse matrix, $\mathbf{S}$ plus a noise matrix $\mathbf{N}$ by $\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N}$ with an algorithm, called GoDec to find desired a low rank matrix $\mathbf{L}$, and a sparse matrix $\mathbf{S}$. In analogy with the LRaSMD model, Chen *et al.* [44] further developed component decomposition analysis (CDA) which represented a data space as a three-component mixture model in orthogonal decomposition given by $\mathbf{X} = \mathbf{PC} + \mathbf{IC} + \mathbf{N}$ where $\mathbf{PC}$, $\mathbf{IC}$, and $\mathbf{N}$ are principal components (PCs), ICs, and noise component, respectively.

B. AD Using DS Coupled With SDPs

Deviating from the approaches described above, this article looks into a new idea of how to implement AD. Statistically speaking, a dataset can be described by two types of statistics, 2OS, and high orders of data statistics (HOS) which can be used to characterize BKG and to specify anomalies, respectively. In other words, instead of using the LRaSMD model, we are more interested in breaking up a data space $\mathbf{X}$ into 2OS-characterized space, $\mathbf{X}_{2\mathrm{OS}}$ and the sphered data space, $\mathbf{X}_{\mathrm{DS}}$. With this assumption, AD can be implemented by two stage processes. One is DS which is designed to remove the 2OS-characterized BKG in the first stage and then followed by extracting anomalies from the sphered data, $\mathbf{X}_{\mathrm{DS}}$ in the second stage. A second new idea is to develop a new concept, called SDP, which is designed to remove noises and interferers in the sphered data space, $\mathbf{X}_{\mathrm{DS}}$.

Although DS has widely been used technique in communications and signal processing, its utility in hyperspectral imaging seems yet to be explored. Compared with many current approaches which intended to use a mixture of Gaussians to estimate, reconstruct, and model BKG, DS allows AD to clean up BKG more effectively other than using Gaussians. As for the idea of SDP, it is originated from the prominent peaks used to select bands by band selection based on a self-mutual information curve [45]. Besides, SDP is also inspired by SC introduced in [43] where SDP converts all the data sample

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values in $\mathbf{X}_{\mathrm{DS}}$ as a probability mass function by normalization to be used as a sparsity density curve of $\mathbf{X}_{\mathrm{DS}}$. Three versions of SDP are derived, SS-SDP, fixed spectral SDP (FS-SDP), and fixed band SDP (FB-SDP) which replace SC used by GoDec in [43]. More specifically, we can create a curve similar to the self-mutual information curve in [45] by plotting all data sample values in the 3-D data cube, $\mathbf{X}_{\mathrm{DS}}$, as a sparsity density curve from which k prominent peaks can be extracted to form an SDP constrained sparse space, $\mathbf{X}_{\mathrm{DS}}^{\mathrm{SDP}}$, from $\mathbf{X}_{\mathrm{DS}}$ in correspondence to k largest nonzero data sample values used by GoDec to form $\mathbf{X}_{\mathrm{GoDec}}^{\mathrm{SC}}$ constructed from its sparse space.

Since BKG is assumed to be characterized by 2OS, it is expected that HOS-specified anomalies will be retained in $\mathbf{X}_{\mathrm{DS}}$. However, it is generally known that hyperspectral noise is non-Gaussian. This indicates that $\mathbf{X}_{\mathrm{DS}}$ may still contain non-Gaussian noises. Therefore, how to extract anomalies from the non-Gaussian noise contaminated $\mathbf{X}_{\mathrm{DS}}$ is a great challenge. To address this issue, two approaches are of particular interest. The first approach is to assume that noise is ubiquitous and cannot be sparse. In this case, either SC or SDP can be used to eliminate noisy data samples from $\mathbf{X}_{\mathrm{DS}}$. However, this also comes with an issue in how to determine the value, k, for SC or SDP.

As a second approach, we implement the well-known HOS statistical signal processing technique, ICA [46] to extract anomalies in its generated ICs from $\mathbf{X}_{\mathrm{DS}}$. Unfortunately, there are three issues associated with ICA. One is that ICA cannot separate anomalies from non-Gaussian noises which are also HOS. Another is that the ICA-generated ICs are not repeatable due to its use of random initial conditions. A third one is that not all ICs are significant. This is because some ICs may contain non-Gaussian noises. Therefore, it is crucial to determine an appropriate number of ICs, j, to be retained with the rest of ICs discarded.

To resolve the issue in determining the values of k and j, Chang [1], Chang and Du [47], Chang [48], and Chang [49] developed an approach using VD in conjunction with the MX-SVD [50] to automatically determine the number of spectrally distinct signatures, p, and the number of BKG signatures m so that the number of rare signal sources to be considered as anomalous signal sources can be estimated by $j = p - m$. As a matter of fact, VD coupled with MX-SVD has been shown to be effective in finding the sparse space $\mathbf{S}$ [44], [51]–[53]. Once the value of j is determined, the number of ICs to be retained is then determined by j and the values of SC and SDP can be also determined by $k = j \times N$ where N is the total number of data pixel vectors.

With the determined k, we can produce an SC/SDP constrained sparse space $\mathbf{X}_{\mathrm{DS}}^{\mathrm{SC/SDP}}$ from $\mathbf{X}_{\mathrm{DS}}$. By using the same value of j, we can also produce an SC/SDP-constrained sparse space $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{SC/SDP}}$ for a j-IC formed component space, $\mathbf{X}_{\mathrm{IC}^j}$, where the FastICA developed [54] can be used to generate the desired j IC from $\mathbf{X}_{\mathrm{DS}}$.

After either $\mathbf{X}_{\mathrm{DS}}^{\mathrm{SDP}}$ without using ICA or $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{SDP}}$ is constructed, AD is then carried out on these spaces by any anomaly detector, such as RX-AD [9], CEM-AD [11], and OSP-AD [52]. However, compared with RX-AD and CEM-AD, it is worth noting that OSP-AD requires prior knowledge of an undesired signature space denoted by $\mathbf{U}$, to be annihilated by OSP. To meet this requirement, GoDec is further implemented on $\mathbf{X}_{\mathrm{DS}}$ to generate the SC or SDP-constrained sparse space, $\mathbf{X}_{\mathrm{DS\text{-}GoDec}}^{\mathrm{SC/SDP}}$ prior to OSP-AD. That is, the GoDec-sparse space produced by imposing SC or SDP gives rise to $\mathbf{X}_{\mathrm{DS\text{-}GoDec}}^{\mathrm{SC/SDP}}$. As a result, imposing SC/SDP on the GoDec-generated sparse space in the original data space $\mathbf{X}$, GoDec-generated sparse space in the sphered space $\mathbf{X}_{\mathrm{DS}}$ and j-IC formed component space, $\mathbf{X}_{\mathrm{IC}^j}$ respectively produced the three difference sparse spaces, $\mathbf{X}_{\mathrm{GoDec}}^{\mathrm{SC/SDP}}$, $\mathbf{X}_{\mathrm{DS\text{-}GoDec}}^{\mathrm{SC/SDP}}$, and $\mathbf{X}_{\mathbf{IC}^j}^{\mathrm{SC/SDP}}$, which can be used for AD.

C. 3-D ROC Curve Analysis for AD Performance

In order to address the performance of AD, BS is an important indicator. The commonly used 2-D ROC curve of detection probability P_{D} versus false alarm probability P_{F} has been shown to be ineffective in [44], [51], [52], [53], [54], and [55] due to the fact that both P_{D} and P_{F} are determined by the same threshold and cannot stand alone to be used as independent criteria. Specifically, when P_{D} is high, P_{F} is also high and vice versa. Consequently, using the area under the curve (AUC) of the 2-D ROC curve of $(P_{\mathrm{D}},P_{\mathrm{F}})$ cannot be used to evaluate BS. To resolve this dilemma, Chang [56] introduced a 3-D ROC curve from which three 2-D ROC curves can be derived, i.e., 2-D ROC curve of $(P_{\mathrm{D}},P_{\mathrm{F}})$, 2-D ROC curve of (P_{D},τ), and 2-D ROC curve of (P_{F},τ). Based on these three 2-D ROC curves, eight detection measures were designed to evaluate AD performance in terms of TD, BS, joint TD and BS (TDBS), and overall detection performance (ODP).

Finally, three popular real hyperspectral image data scenes were used for experiments to demonstrate the superior BS suppressibility of the proposed IC-SDP-AD operating in $\mathbf{X}_{\mathrm{DS}}$ compared with most recently developed AD methods, RPCA in [34], LSDM-MoG in [27], and PTA in [42], all of which did not separate noise to be an independent subspace as LRaSMD model does. As a result, these methods have been shown to perform poorly in [44]. In addition, IC-SDP-AD was also shown to be competitive and perform slightly better than CDA in [44] and OSP-AD in [52]. Specifically, details of experimental results using 3-D ROC-derived detection measures will be discussed in Sections VII and VIII.

The main contributions of this article are summarized as follows.

1) It introduces DS to remove BKG instead of using low rank space to represent BKG.
2) It applies FastICA to the sphered data space, $\mathbf{X}_{\mathrm{DS}}$, to generate a j-IC component space that can be used to form a sparse space, $\mathbf{X}_{\mathrm{IC}^j}$.
3) It develops a new concept, SDP with three different versions, SS-SDP, FS-SDP, and FB-SDP.
4) It imposes SC/SDP on $\mathbf{X}_{\mathrm{DS}}$ and $\mathbf{X}_{\mathrm{IC}^j}$ to produce various SC/SDP-constrained spaces, $\mathbf{X}_{\mathrm{GoDec}}^{\mathrm{SC/SDP}}$, $\mathbf{X}_{\mathrm{IC}^j}^{\mathrm{SC/SDP}}$, and $\mathbf{X}_{\mathrm{DS\text{-}GoDec}}^{\mathrm{SC/SDP}}$ for AD.
5) It conducts comparative studies and analyses among the latest developed AD methods to demonstrate that using SC/SDP can significantly improve anomaly detectability.

The remainder of this article is organized as follows. Section II briefly reviews GoDec and its algorithm.

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Section III is devoted to the new concept of SDP with three different versions. Section IV describes methods to determine the number of ICs. Section V develops DS-SDP and DS-IC-SDP approaches for two ADs, RX-AD, and OSP-AD. Section VI includes 3-D ROC curve-derived detection measures to evaluate AD performance. Section VII describes images to be used for experiments and performs image experiments along with experimental discussions. Section VIII presents a comparative analysis among some latest developed LRaSR-based AD methods. Section IX draws some conclusions.

II. Go Decomposition (GoDEC)

Over the past years, quite a few works on AD have been reported in the literature. A most recent brief review can be found in [3]. This section only reviews LRaSMD whose introduced SC is the key to develop our new developed SDP in Section III.

A. Go Decomposition (GoDEC)

LRaSMD models a data matrix $\mathbf{X}$ as a linear decomposition of a low-rank matrix, $\mathbf{L}$, and a sparse matrix, $\mathbf{S}$, plus a noise matrix $\mathbf{N}$, represented by

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N}. \tag{1}$$

In order to produce (1), GoDec [43] was developed to find the low-rank matrix $\mathbf{L}$ and the sparse matrix $\mathbf{S}$ by solving the following constrained problem:

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \text{subjcet to } \text{rank}(\mathbf{L}) \le r, \quad \text{card}(\mathbf{S}) \le k \tag{2}$$

where $\text{card}(\mathbf{S}) = SC$. Zhang *et al.* [57] suggested that the value of k be $k = rN$ without explanation. In [51], the model (2) has been shown to perform better than the two-component decomposition, $\mathbf{X} = \mathbf{L} + \mathbf{S}$, produced by RPCA.

The idea of GoDec [43] is to use bilateral random projection (BRP) to solve the following low-rank approximation:

$$\mathbf{L} = \mathbf{Q}_1 \left[\mathbf{R}_1 \left(\mathbf{A}_2^T \mathbf{Y}_1 \right)^{-1} \mathbf{R}_2^T \right]^{1/(2q+1)} \mathbf{Q}_2^T \tag{3}$$

where $\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{R}_1, \mathbf{R}_2$ are QR decomposition of $\mathbf{Y}_1, \mathbf{Y}_2$, and q is an empirically predetermined parameter. In addition, the model in (2) can be further solved by minimizing the following decomposition error:

$$\min_{\mathbf{L}_{N\times L}, \mathbf{S}_{N\times L}} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L} - \mathbf{S}_{N\times L}||_F^2 \tag{4}$$

$$\text{subject to } \text{rank}(\mathbf{L}_{N\times L}) \le m \text{ and } \text{card}(\mathbf{S}_{N\times L}) \le jN \tag{5}$$

where $||\cdot||_F$ is the Frobenius matrix norm. Since (2) does not have an analytical solution, it must rely on a numerical algorithm, which is GoDec to find solutions by the following iterative processes:

$$\mathbf{L}_{N\times L}^{(t)} = \arg \min_{\text{rank}(\mathbf{L}_{N\times L}) \le m} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L} - \mathbf{S}_{N\times L}^{(t-1)}||_F^2 \tag{6}$$

$$\mathbf{S}_{N\times L}^{(t)} = \arg \min_{\text{card}(\mathbf{S}_{N\times L}) \le jN} ||\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L}^{(t-1)} - \mathbf{S}_{N\times L}||_F^2. \tag{7}$$

Specifically, (6) and (7) can be solved iteratively by

$$\mathbf{L}_{N\times L}^{(t)} = \mathbf{Q}_1 [\mathbf{R}_1 (\mathbf{A}_2^T \mathbf{Y}_1)^{-1} (\mathbf{R}_2)^T]^{1/(2q+1)} \mathbf{Q}_2^T \tag{8}$$

$$\mathbf{S}_{N\times L}^{(t)} = P_\Omega (\mathbf{X}_{N\times L} - \mathbf{L}_{N\times L}^{(t)}) \tag{9}$$

where $P_\Omega(\cdot)$ is the projection of a matrix onto an entry set Ω and was discussed in great detail in [44]. For more details about GoDec, we refer to https://sites.google.com/site/godecomposition/home.

III. Sparsity Density Peaks (SDPs)

As mentioned in Section II, $\text{card}(\mathbf{S}) = \text{SC} = k$ is a unique property that was overlooked in many LRaSMD-based methods. It is quite different from the sparsity order commonly used by LRaSR, which is generally obtained by regularization-constrained optimization problems through Lagrange multipliers. Interestingly, the great potential of SC in AD has not been explored until [51]. Since then, several approaches have used SC to show its success in AD [44], [51]–[53].

How to select the k largest nonzero entries in $P_\Omega(\mathbf{S})$ in (9) is indeed a key issue to the success of AD in [44], [51], [52], and [53]. If we interpret all entries in $P_\Omega(\mathbf{S})$ which are prioritized by their magnitudes in descending order as bands which are also prioritized according to a band prioritization criterion, then selecting the k largest nonzero entries in $P_\Omega(\mathbf{S})$ by SC can also be viewed as selecting bands with the k highest priorities by band selection. Therefore, the issue of interband correlation arising in band prioritization will also be applied to SC as well. That is, among the k largest nonzero entries in $P_\Omega(\mathbf{S})$ selected by SC, the intersample spectral and spatial correlation may also exist and cause sample redundancy.

To resolve this issue, the prominent peaks used in band selection in [45] inspire the development of a new concept, called, SDP. Its idea is to convert all data sample values in the sparse space, $\mathbf{S}$ to a probability vector by normalization from which a sparsity density curve can be plotted according to their probabilities. It then follows a similar approach proposed in [45] to find prominent peaks from this sparsity density curve. Three versions of SDP can be derived according to spectral, spatial and joint SS correlation as follows.

A. SS-SDP of a Spatial–Spectral Data Cube

Assume that N_r, N_c, and L are the number of rows, number of columns, and number of spectral bands, and k is given the number of peaks to be found, $k = n_{\text{peaks}}$. Let $\mathbf{r}_{(n_r,n_c)}^{\text{SS}} = (r_{(n_r,n_c),1}^{\text{SS}}, r_{(n_r,n_c),2}^{\text{SS}}, \ldots, r_{(n_r,n_c),L}^{\text{SS}})^T$ be a spectral sample vector at spatial location (n_r, n_c) in $\mathbf{S}$ where $r_{(n_r,n_c),l}^{\text{SS}}$ can be considered as an SS data sample value at spatial location (n_r, n_c) and the lth-band in $\mathbf{S}$. Therefore, there are a total of $N \times L$ SS data sample values in $\mathbf{S}$.

We now plot $\left\{r_{(n_r,n_c),l}^{\text{SS}}\right\}_{n_r,n_c,l}^{N_r,N_c,L}$ in $\mathbf{S}$ as a curve and find $n_{\text{peaks}} = k$ prominent peaks, denoted by

$$\Omega_{\text{peaks}}^{\text{SS-SDP}} = \left\{ r_i^{\text{peak}} \right\}_{i=1}^{n_{\text{peaks}}}. \tag{10}$$

Following a similar idea to $\text{SC} = k$ used in GoDec, we can find an SS-SDP space from all data samples $\{r_{(n_r,n_c),l}^{\text{SS}}\}_{n_r,n_c,l}^{N_r,N_c,L}$ in $\mathbf{S}$, $\mathbf{X}^{\text{SS-SDP}}$, which is comprised of new data samples by imposing SS-SDP on $\mathbf{S}$ as follows:

$$r_{(n_r,n_c),l}^{\text{SS-SDP}} = \begin{cases} r_{(n_r,n_c),l}^{\text{SS}}, & \text{if } r_{(n_r,n_c),l}^{\text{SS}} \in \Omega_{\text{peaks}}^{\text{SS-SDP}} \\ 0, & \text{if } r_{(n_r,n_c),l}^{\text{SS}} \notin \Omega_{\text{peaks}}^{\text{SS-SDP}} \end{cases} \tag{11}$$

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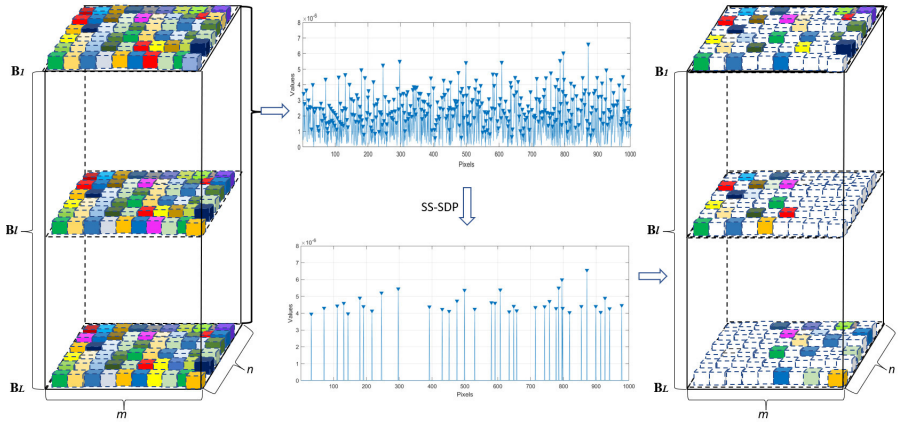


Fig. 1. SS-SDP of an SS data cube. (a) Spatial–spectral **S** data cube. (b) Spatial–spectral density peaks with $k = j \times m \times n$ with $j = 6$. (c) SS-SDP of a spatial–spectral data cube.

where $r^{\text{SS}}_{(n_r,n_c),l}$ runs through the entire **S** cube, i.e., $\{r^{\text{SS}}_{(n_r,n_c),l}\}^{N_r,N_c,L}_{n_r,n_c,l}$. By virtual of (11), we can derive an SS peak vector, $\mathbf{r}^{\text{SS-SDP}}_{(n_r,n_c)}$, at spatial location (n_r, n_c) in **S** by

$$\mathbf{r}^{\text{SS-SDP}}_{(n_r,n_c)} = \left(r^{\text{SS-SDP}}_{(n_r,n_c),1}, r^{\text{SS-SDP}}_{(n_r,n_c),2}, \ldots, r^{\text{SS-SDP}}_{(n_r,n_c),L}\right)^T \tag{12}$$

which makes up $\mathbf{X}^{\text{SS-SDP}}$. The algorithm for finding SDPs is summarized in Algorithm 1 as below.

Algorithm 1 Finding $\mathbf{X}^{\text{SS-SDP}}$ From **S**

1 Initial condition: $n_{\text{peaks}} = k$.
2 Find the prominent spatial sparsity density peaks of **S** $\mathbf{X}^{\text{SS-SDP}} = \{r^{\text{SS-SDP}}_{(n_r,n_c),l}\}^{N_r,N_c,L}_{n=1,n_c=1,l=1}$ by the following procedure
   a. Extend a horizontal line from the point $r^{\text{SS}}_{(n_r,n_c),l}$ to the left and the right until the line either
      - crosses the curve because there is a higher point or
      - reaches the left or right end of the curve
   b. Find the minimum point in each of the two intervals defined in step 2(a).
   c. Use the higher point of the two interval minima as the reference level. The prominence is defined as the height of $r^{\text{SS}}_{(n_r,n_c),l}$ above the reference level.
3 Then the peaks of **S** are defined by $\Omega^{\text{SS-SDP}}_{peaks} = \{r^{\text{peak}}_i\}^{n_{\text{peaks}}}_{i=1}$ by (10)
4 Find $\mathbf{X}^{\text{SS-SDP}} = \{\mathbf{r}^{\text{SS-SDP}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r,n_c}$ where $\mathbf{r}^{\text{SS-SDP}}_{(n_r,n_c)}$ defined by (12)

Fig. 1 depicts how SS-SDP is used to construct $\mathbf{X}^{\text{SS-SDP}}$. Fig. 1(a) shows original SS values in $\{r^{\text{SS}}_{(n_r,n_c),l}\}^{N_r,N_c,L}_{n_r,n_c,l}$ in an **S** data cube where different colors represent different values of $r^{\text{SS}}_{(n_r,n_c),l}$ spatial location (n_r, n_c) and the lth-band in **S**. Fig. 1(b) plots all $\{r^{\text{SS}}_{(n_r,n_c),l}\}^{N_r,N_c,L}_{n_r,n_c,l}$ as a curve from which $k = j \times m \times n$ are found where $N = m \times n$. Fig. 1(c) shows the final construction of $\mathbf{X}^{\text{SS-SDP}}$ where the blank pixels indicate zero values.

B. j-FS-SDP of a Spectral Pixel

In Section III-A, SS-SDP finds prominent peaks from the curve plotted by all the data sample values $\{r^{\text{SS}}_{(n_r,n_c),l}\}^{N_r,N_c,L}_{n_r,n_c,l}$ in an 3-D data cube **S** by setting the number of peaks, $n_{\text{peaks}} = k$, in correspondence to SC $= k = j \times N$ used in GoDec. Since k is determined by j which is the dimensionality of **S**, we can assume that each spectral pixel vector has a sparse representation with its sparsity order specified by j defined in the same way as many LRaSR-based models do. With this in mind, we can derive a j-FS-SDP (FS-SDP) for each spectral pixel $\mathbf{r}^{\text{spectral}}_{(n_r,n_c)}$ at the spatial location (n_r, n_c).

Let $\mathbf{\Omega}_{\text{peaks}}$ be

$$\mathbf{\Omega}_{\text{peaks}} = \Omega^j_{(n_r,n_c)} = \left\{r^{\text{peak}}_{(n_r,n_c),i}\right\}^j_{i=1} \tag{13}$$

with $n_{\text{peaks}} = j$ found spectral peaks for $\mathbf{r}^{\text{spectral}}_{(n_r,n_c)}$. Then, we construct

$$\mathbf{r}^{j-\text{FS-SDP}}_{(n_r,n_c)} = \left(r^{j-\text{FS-SDP}}_{(n_r,n_c),1}, r^{j-\text{FS-SDP}}_{(n_r,n_c),2}, \ldots, r^{j-\text{FS-SDP}}_{(n_r,n_c),L}\right)^T \tag{14}$$

where $r^{j-\text{FS-DP}}_{(n_r,n_c),l}$ is defined by

$$r^{j-\text{FS-SDP}}_{(n_r,n_c),l} = \begin{cases} r^{\text{spectral}}_{(n_r,n_c),l}, & \text{if } r^{\text{spectral}}_{(n_r,n_c),l} \in \Omega^j_{(n_r,n_c)} \\ 0, & \text{if } r^{\text{spectral}}_{(n_r,n_c),l} \notin \Omega^j_{(n_r,n_c)}. \end{cases} \tag{15}$$

Since there are no more than j nonzero elements in $\Omega^j_{(n_r,n_c)}$ in (13) for each spectral vector $\mathbf{r}^{\text{spectral}}_{(n_r,n_c)}$, $||\mathbf{r}^{j-\text{FS-SDP}}_{(n_r,n_c)}||_{l_0} \leq j$, where $||\cdot||_{l_0}$ is defined as l_0-norm of $\mathbf{r}^{j-\text{FSDP}}_{(n_r,n_c)}$.

Similar to Algorithm 1, we can also develop an algorithm for finding $\mathbf{X}^{j-\text{FS-SDP}}$ with $n_{\text{peaks}} = j$ as Algorithm 2.

Algorithm 2 Finding $\mathbf{X}^{j-\text{FS-SDP}}$ From **S**

1 Initial condition: $n_{\text{peaks}} = j$.
2 Find the prominent spatial sparsity density peaks of **S**
3 The peaks of **S** are defined by $\Omega^j_{(n_r,n_c)} = \left\{r^{\text{peak}}_{(n_r,n_c),i}\right\}^j_{i=1}$.
4 Find $\mathbf{X}^{j-\text{FS-SDP}} = \{\mathbf{r}^{j-\text{FS-SDP}}_{(n_r,n_c)}\}^{N_r,N_c}_{n_r,n_c}$ where $\mathbf{r}^{j-\text{FS-SDP}}_{(n_r,n_c)}$ defined by (14)

Fig. 2 depicts how j-FS-SDP is used to construct $\mathbf{X}^{j-\text{FS-SDP}}$ where Fig. 2(a) shows spectral values of $\mathbf{r}^{\text{spectral}}_{(n_r,n_c)}$ at the spatial

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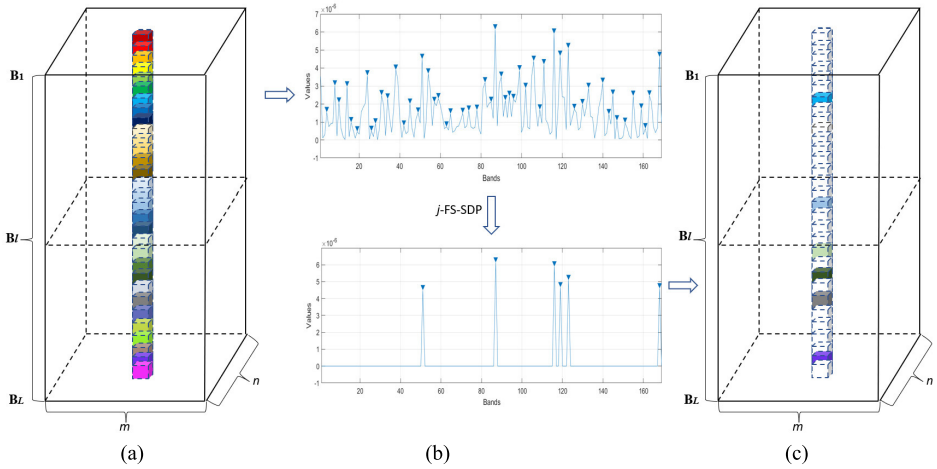


Fig. 2. j-FS-DSP of a spectral pixel at (m, n). (a) Spectral pixel at (m, n) in S. (b) j-FS-SDPs with $j = 6$. (c) j-FS-SDP of a spectral pixel.

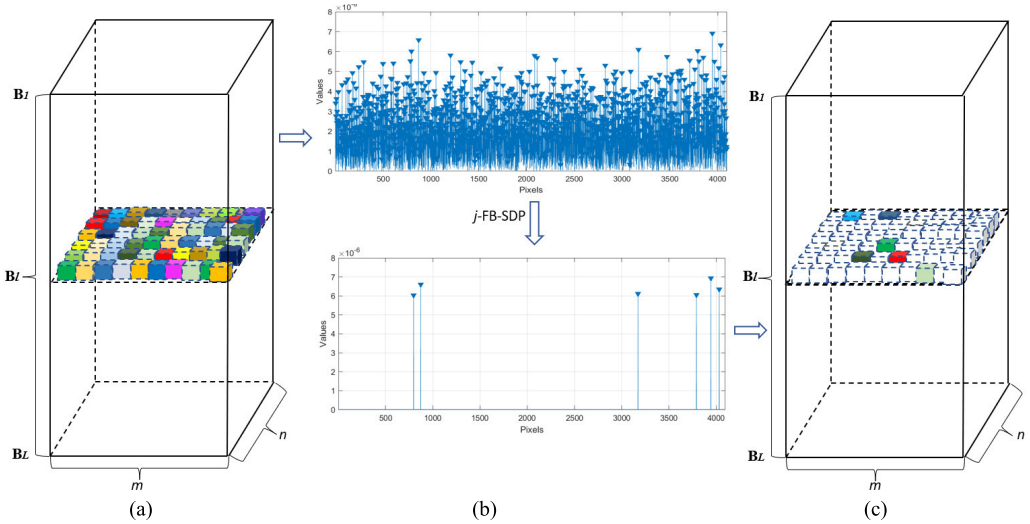


Fig. 3. j-FB-SDP of the lth band image, Bl, in S. (a) lth-band image, Bl, in S. (b) j-FB-SDPs with $j = 6$. (c) j-FB-SDP of Bl.

location (m, n) in $\mathbf{S}$ and Fig. 2(b) plots all $\{r^{\text{spectral}}_{(n_r,n_c),l}\}_{l=1}^{L}$ as a curve from which $n_{\text{peaks}} = j$ peaks are extracted. Fig. 2(c) shows the final construction of $\mathbf{X}^{j-\text{FS-SDP}}$ where the blank pixels indicate zero values.

C. j-FB-SDP of a Band Image

As an alternative to j-FS-SDP which finds peaks from a spectral pixel vector, $\mathbf{r}^{\text{spectral}}_{(n_r,n_c)}$ at the spatial location (n_r, n_c), we can also find spatial peaks from each of band images. Let $\mathbf{B}_l$ be the lth-band image denoted by $\mathbf{B}^{\text{S}}_l = [r^{\text{S}}_{(n_r,n_c),l}]_{n=1,n_c=1}^{N_r,N_c}$ and Ω_{peaks} be $\Omega_{\text{peaks}} = \Omega^j_{\mathbf{B}_l} = \{r^{\text{peak}}_i\}_{i=1}^{j}$. They derive a j-FB-SDP (j-FB-SDP) for each band image as follows:

$$r^{j-\text{FBDP}}_{(n_r,n_c),l} = \begin{cases} r^{\text{S}}_{(n_r,n_c),l}, & \text{if } r^{\text{S}}_{(n_r,n_c),l} \in \Omega^j_{\mathbf{B}_l} \\ 0, & \text{if } r^{\text{S}}_{(n_r,n_c),l} \notin \Omega^j_{\mathbf{B}_l}. \end{cases} \tag{16}$$

Then, we can construct $\mathbf{B}^{j-\text{FB-SDP}}_l = [r^{j-\text{FB-SDP}}_{(n_r,n_c),l}]_{n=1,n_c=1}^{N_r,N_c}$ for a fixed band image $\mathbf{B}_l$ with an algorithm described as Algorithm 3.

Algorithm 3 Finding $\mathbf{X}^{j-\text{FB-SDP}}$ From $\mathbf{S}$

1 Initial condition: $n_{\text{peaks}} = j$.
2 Find the prominent spatial sparsity density peaks of $\mathbf{S}$
3 The peaks of $\mathbf{S}$ are defined by
$\Omega_{\text{peaks}} = \Omega^j_{\mathbf{B}_l} = \left\{r^{\text{peak}}_i\right\}_{i=1}^{j}$ by (13)
4 Find $\mathbf{X}^{j-\text{FB-SDP}} = \left\{\mathbf{B}^{\text{FB-SDP}}_l\right\}_l^L$ where $r^{j-\text{FB-SDP}}_{(n_r,n_c),l}$ defined by (16)

Fig. 3 depicts how j-FB-SDP is used to construct $\mathbf{X}^{j-\text{FB-SDP}}$ where Fig. 3(a) shows spatial values of $\mathbf{B}^{\text{S}}_l$ at the lth-band in $\mathbf{S}$, and Fig. 3(b) plots all $\{r^{\text{S}}_{(n_r,n_c),l}\}_{n=1,n_c=1}^{N_r,N_c}$ as a curve from which $n_{\text{peaks}} = j$ peaks are found. Fig. 3(c) shows the final construction of $\mathbf{X}^{j-\text{FB-SDP}}$ where the blank pixels indicate zero values.

It should be noted that for the sake of simplicity, $\mathbf{S}$ presented in this section represents a generic sparse space. For example, when GoDec is applied to the original data space $\mathbf{X}$, the sparse space is specified by $\mathbf{S}$ as defined in (2). Specifically, when $\mathbf{S}$ is the sphered data space, then $\mathbf{S} = \mathbf{X}_{\text{DS}}$. When GoDec is applied

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to sphered dada, $\mathbf{X}_{\mathrm{DS}}$, then $\mathbf{S} = \mathbf{X}^{\mathbf{S}}_{\text{DS-GoDec}}$. In addition, when FastICA is applied to $\mathbf{X}_{\mathrm{DS}}$ to find a j-IC component space, $\mathbf{S} = \mathbf{X}_{\mathrm{IC}^j}$.

Finally, it is interesting to note that by virtue of SDP, the CDA model developed in [44] can also be extended to CDA-SS-SDP, CDA-j-FS-SDP, and CDA-j-FB-SDP in correspondence to IC-SS-SDP, IC-j-FS-SDP, and IC-j-FB-SDP, respectively.

IV. Determination of Number of ICs and Value of SDP

As noted in Sections II and III, the core of SC/SDP is the value of j, which determines the value of SC $= k$ and the number of ICs to be retained in SDP.

If we consider anomalies as rare signal sources, an approach developed by Kuybeda *et al.* [50], called MX-SVD, can be used to determine the rank of a rare signal subspace within the signal subspace whose rank, p, can be determined by the maximum orthogonal complement algorithm (MOCA). Unfortunately, it has been shown in [58] and [59] that MOCA was the maximum likelihood detector and did not perform well in estimating the value of p compared to the Neyman–Pearson detector used in estimating VD. Therefore, in this article, VD is used to replace MOCA to determine p. In other words, the rank of BKG subspace + rare signal subspace (BKG + RSS) can be estimated by VD as

$$\mathrm{rank}^{\mathrm{VD}}_{\mathrm{BKG+RSS}} = p. \tag{17}$$

Once the value of p is determined, MX-SVD is then used to determine the value of j. Its step-by-step implementation is summarized in the following.

Algorithm 4 MX-SVD

1 Initial condition: p determined by (17). Set $h = 0$. Let $\Omega_0 = \varnothing$ be the initial rare signal subspace and $\mathbf{\Psi}_p = [\boldsymbol{\psi}_1 \boldsymbol{\psi}_2 \cdots \boldsymbol{\psi}_p]$ be the initial BKG subspace where $\boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \ldots, \boldsymbol{\psi}_p$ are principal left singular vectors of $\mathbf{X}$.

2 For the hth signal subspace, $[\mathbf{\Psi}_{p-h}|\mathbf{\Omega}_h]$, we find $(h+1)^{\mathrm{st}}$ rare signal source given by

$$\mathbf{t}_{h+1} = \arg\max_{\mathbf{r}\in\mathbf{X}} ||P^{\perp}_{[\mathbf{\Psi}_{p-h}|\mathbf{\Omega}_h]}\mathbf{r}|| \tag{18}$$

$$\omega_{h+1} = ||P^{\perp}_{[\mathbf{\Psi}_{p-h}|\mathbf{\Omega}_h]}\mathbf{t}_{h+1}|| \tag{19}$$

3 Form a new rare signal subspace $\mathbf{\Omega}_{h+1}$ given by

$$\mathbf{\Omega}_{h+1} = [\mathbf{\Omega}_h|\mathbf{t}_{h+1}] \tag{20}$$

4 Find $p-h-1$ principal left singular vectors of $P^{\perp}_{[\mathbf{\Omega}_{h+1}]}\mathbf{X}$, $\boldsymbol{\psi}_1, \boldsymbol{\psi}_2, \ldots, \boldsymbol{\psi}_{p-h-1}$ to form a new BKG subspace $\mathbf{\Psi}_{p-h-1}$ given by

$$\mathbf{\Psi}_{p-h-1} = [\boldsymbol{\psi}_1 \boldsymbol{\psi}_2 \cdots \boldsymbol{\psi}_{p-h-1}] \text{ and } [\mathbf{\Psi}_{p-h-1}|\mathbf{\Omega}_{h+1}] \tag{21}$$

5 Check if $h+1 = p$. If yes, go to step 6. Otherwise, go to step 2

V. Data Sphering-Sparsity Density Peak-Anomaly Detection

This section uses the three types of As, RX-AD [4], CEM-AD [11], and OSP-AD [52] to illustrate how DS-SDP-AD is designed. According to [51], these three ADs can be implemented in two separate individual spaces, $\mathbf{A}$ and $\mathbf{B}$, which indicate sparse and BKG spaces, respectively, as follows:

$$\delta^{\text{RX-AD}}_{\mathrm{DS}}(\mathbf{r}) = (\mathbf{r}_{\mathbf{A}} - \mu_{\mathbf{A}})^T \mathbf{K}^{-1}_{\mathbf{B}} (\mathbf{r}_{\mathbf{A}} - \mu_{\mathbf{A}}) \tag{22}$$

$$\delta^{\text{CEM-AD}}_{\mathrm{DS}}(\mathbf{r}) = \kappa \mathbf{r}^T_{\mathbf{A}} \mathbf{R}^{-1}_{\mathbf{B}} \mathbf{r}_{\mathbf{A}} \quad \text{with } \kappa = 1 \tag{23}$$

$$\delta^{\text{OSP-AD}}_{\mathrm{DS}}(\mathbf{r}) = \kappa \mathbf{r}^T_{\mathbf{A}} P^{\perp}_{\mathbf{B}} \mathbf{r}_{\mathbf{A}} \quad \text{with } \kappa = 1 \tag{24}$$

where $\mathbf{A}$ and $\mathbf{B}$ are specified to enhance AD and BKG suppression/removal, respectively.

A. Anomaly Detector Operated in the Data Sphered Space Using SC/SDP (DS-SC/SDP-AD)

Since a large part of the BKG is supposed to be removed in $\mathbf{X}_{\mathrm{DS}}$, we can find SDP directly from $\mathbf{X}_{\mathrm{DS}}$ to generate an SDP space, $\mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}}$, and then further design RX-AD, CEM-AD, and OSP-AD with $\mathbf{A} = \mathbf{B} = \mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}}$ as follows:

$$\delta^{\text{RX-AD}}_{\text{DS-SDP}}(\mathbf{r}) = (\mathbf{r} - \mu)^T \mathbf{K}^{-1} (\mathbf{r} - \mu) \tag{25}$$

$$\delta^{\text{CEM-AD}}_{\text{DS-SC/SDP}}(\mathbf{r}) = \kappa \mathbf{r}^T \mathbf{R}^{-1} \mathbf{r} \tag{26}$$

$$\delta^{\text{OSP-AD}}_{\text{DS-SC/SDP}}(\mathbf{r}) = \kappa \mathbf{r}^T_{\mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}}} P^{\perp}_{(\mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}})^c} \mathbf{r}_{\mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}}} \tag{27}$$

where $\mathbf{U} = (\mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}})^c = \mathbf{X}_{\mathrm{DS}} - \mathbf{X}^{\mathrm{SC/SDP}}_{\mathrm{DS}}$.

Although BKG can be removed via DS, it is often the case that some BKG residuals may be leaked into $\mathbf{X}_{\mathrm{DS}}$. Therefore, in the following, two approaches are further implemented in $\mathbf{X}_{\mathrm{DS}}$ to deal with this issue.

B. Anomaly Detector Operated in the GoDec Using SDP (DS-GoDec-SDP-AD)

Unlike the GoDec implemented directly in the original data space $\mathbf{X}$, this section derives a DS-GoDec-SDP approach which implements GoDec in $\mathbf{X}_{\mathrm{DS}}$ to generate the low rank and sparse spaces, denoted by $\mathbf{X}^{\mathbf{L}}_{\text{Ds-GoDec}}$ and $\mathbf{X}^{\mathbf{S}}_{\text{Ds-GoDec}}$, respectively. In other words, DS-GoDec-SDP uses GoDec to generate the low rank space in $\mathbf{X}_{\mathrm{DS}}$ to further remove BKG residuals, while generating a sparse space in $\mathbf{X}_{\mathrm{DS}}$ to enhance the detection of anomalies. In this case, the three ADs in (25)–(27) can be implemented by letting $\mathbf{B} = \mathbf{X}^{\mathbf{L}\text{-SC/SDP}}_{\text{Ds-GoDec}}$ and $\mathbf{A} = \mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{Ds-GoDec}}$ as follows:

$$\delta^{\text{RX-AD}}_{\text{DS-GoDec-SC/SDP}}(\mathbf{r}) = \left(\mathbf{r}_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} - \mu_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}}\right)^T \times \mathbf{K}^{-1}_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} \left(\mathbf{r}_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} - \mu_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}}\right) \tag{28}$$

$$\delta^{\text{CEM-AD}}_{\text{DS-GoDec-SDP}}(\mathbf{r}) = \mathbf{r}^T_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} \mathbf{R}^{-1}_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} \mathbf{r}_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} \tag{29}$$

$$\delta^{\text{OSP-AD}}_{\text{DS-GoDec}}(\mathbf{r}) = \mathbf{r}^T_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}} P^{\perp}_{\mathbf{X}^{\mathbf{L}}_{\text{DS-GoDec}}} \mathbf{r}_{\mathbf{X}^{\mathbf{S}\text{-SC/SDP}}_{\text{DS-GoDec}}}. \tag{30}$$

C. Anomaly Detector Operated in the Data-Sphered IC Space Using SDP (IC-SDP-AD)

As noted earlier in the introduction, prior to FastICA, DS must be performed as a preprocessing step because IC requires DS as a preprocessing of removing 2OS. Therefore, DS-IC is abbreviated as IC for simplicity. When FastICA is implemented, three key issues must be resolved. One is a random initial condition used by FastICA. Another is how

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TABLE I
COMPARISON OF KEY DIFFERENCES AMONG LOW RANK AND SPARSE SPACE-BASED MODELS FOR AD

method	model	BKG space	anomaly space	anomaly detector
LRR [40]	$\min_{\mathbf{S},\mathbf{E}}\left\{\text{rank}(\mathbf{S})+\lambda \parallel \mathbf{E} \parallel_{2,1}\right\}$ subject to $\mathbf{X}=\mathbf{DS}+\mathbf{E}$	**D**	**E**	RX-AD
RPCA [34]/RSL [38]	$\mathbf{X}=\mathbf{L}+\mathbf{S}$ / $\mathbf{X}=\mathbf{S}+\mathbf{LR}$	**L/LR**	**S**	RX-AD
GoDec [43]	$\mathbf{X}=\mathbf{L}+\mathbf{S}+\mathbf{N}$	**L/L+S**	**L/L+S**	$\mathbf{r}^T_{\mathbf{X}_{\mathbf{L/L+S}}}\mathbf{R}^{-1}_{\mathbf{X}_{\mathbf{L/L+S}}}\mathbf{r}_{\mathbf{X}^{\text{SC}}_{\mathbf{L/L+S}}}$
OSP-AD [52]	$\mathbf{X}=\mathbf{L}+\mathbf{S}+\mathbf{N}$	$\left(\mathbf{X}^{\text{SC/SDP}}_{\mathbf{IC}^j}\right)^c$	$\mathbf{X}^{\text{SC/SDP}}_{\mathbf{IC}^j}$	$\mathbf{r}^T_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}P^{\perp}_{\left(\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}\right)^c}\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}$
CDA [44]	$\mathbf{X}=\mathbf{PC}+\mathbf{IC}+\mathbf{N}$	$\mathbf{X}_{\mathbf{PC}}$	$\mathbf{X}^{\text{SC}}_{\mathbf{IC}^j}$	$\mathbf{r}^T_{\mathbf{X}^{\text{SC/SDP}}_{\text{CDAIC}}}\mathbf{R}^{-1}_{\mathbf{X}^{\text{SC/SDP}}_{\text{CDAIC}}}\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{CDAIC}}}$
CEM [11]	$\mathbf{X}=\mathbf{L}+\mathbf{S}+\mathbf{N}$	$\mathbf{X}_{\mathbf{L}}$	$\mathbf{X}^{\text{SC/SDP}}$	$\mathbf{r}^T_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\mathbf{R}^{-1}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}$
IC-SC/SDP-AD in Section V.C	DS	2OS	$\mathbf{X}^{\text{SC/SDP}}_{\text{IC}^j}$	$\mathbf{r}^T_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\mathbf{R}^{-1}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}$

many ICs needed to be generated to form a sparse space for AD. A third one is how to deal with the non-Gaussian noise issue because ICA cannot differentiate anomalies from non-Gaussian noises.

1) As for the first issue, a total of L ICs can be generated by FastICA. However, there are some issues in implementing FastICA. First of all, FastICA-generated ICs are not necessarily in order of information significance as the way that PCs are generated by PCA in accordance with decreasing magnitude of eigenvalues. Another is that ICs generated by FastICA in different runs do not necessarily appear in the same order. That is, ICs are generally not reproducible due to its use of random initial conditions. Therefore, an IC generated earlier by FastICA is not necessarily more significant than one generated later. In order to resolve the issue in the use of the random initial projection unit vectors by FastICA, an approach is to use the ATGP developed in [60] to generate a specific set of initial projection vectors for each IC. Then, FastICA utilizes these ATGP-generated initial projection vectors to initialize FastICA for each of ICs as described below.

Algorithm 5 ATGP-FastICA

1 Step 1: Apply the ATGP to $\mathbf{X}_{\text{DS}}$ to find j target pixel vector, $\{\mathbf{t}_i\}_i^j$ used as initial projection vectors, one for each IC.
2 Step 2: Use FastICA to find j independent components, $\{\text{IC}_i\}_{i=1}^j$ where the ith IC_i is generated by FastICA with initial projection vectors specified by the ith ATGP-generated target pixel vectors obtained in step 1

2) Regarding the second issue, the value of j determined by VD/MXSVD in Section IV can be used to determine how many ICs should be used to form a sparse space. Since ATGP-FastICA has prioritized all the ICs, we can simply take the first j ATGP-FastICA generated ICs to form a j-IC component space, $\mathbf{X}_{\text{IC}^j}$.

3) Finally, to address the third issue, the SC/SDP developed in Section III can be used for this purpose to find an SDP-constrained sparse space, $\mathbf{X}^{\text{SC/SDP}}_{\text{IC}^j}$, from $\mathbf{X}_{\text{IC}^j}$.

Integrating the above approaches into one IC-SDP-based AD algorithm yields the following IC-SDP-AD, Algorithm 6.

Algorithm 6 IC-SC/SDP-AD

1 Initial conditions: j is given.
2 Apply ATGP-FastICA to $\mathbf{X}_{\text{DS}}$ to find the first j IC vectors $\{\mathbf{v}_s\}_{s=1}^j$ and form $\mathbf{IC}^j=\left[\mathbf{v}_1\mathbf{v}_2\cdots\mathbf{v}_j\right]$.
3 Use (5) to find $\mathbf{X}_{\text{IC}^j}=P_{\text{IC}^j}\mathbf{X}_{\text{DS}}$ where

$$P_{\text{IC}^j}=\mathbf{IC}^j\left(\left(\mathbf{IC}^j\right)^T\mathbf{IC}^j\right)^{-1}\left(\mathbf{IC}^j\right)^T. \tag{31}$$

4 Determine k for SC/SDP. Let $k=j\times N$ be the number of sparsity peaks.
5 Plot the value of data samples in the data cube, $\mathbf{X}_{\text{IC}^j}$ and then find its k prominent peaks denoted by $\mathbf{X}^{\text{SC/SDP}}_{\text{IC}^j}$. Three different versions to find SDP developed in Section III can be used for this purpose.

For the sake of simplicity, we can design RX-AD, CEM-AD, and OSP-AD where $\mathbf{A}=\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}$ and $\mathbf{B}=(\mathbf{X}^{\text{SC/SDP}}_{\text{IC}})^c$

$$\delta^{\text{RX-AD}}_{\text{IC-SC/SDP}}(\mathbf{r})=\left(\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}-\mu_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\right)^T\mathbf{K}^{-1}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\left(\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}-\mu_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\right) \tag{32}$$

$$\delta^{\text{CEM-AD}}_{\text{IC-SC/SDP}}(\mathbf{r})=\mathbf{r}^T_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\mathbf{R}^{-1}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}} \tag{33}$$

$$\delta^{\text{OSP-AD}}_{\text{IC-SDP}}(\mathbf{r})=\mathbf{r}^T_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}}P^{\perp}_{(\mathbf{X}^{\text{SC/SDP}}_{\text{IC}})^c}\mathbf{r}_{\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}} \tag{34}$$

where $\mathbf{U}=(\mathbf{X}^{\text{SC/SDP}}_{\text{IC}})^c=\mathbf{X}_{\text{IC}}-\mathbf{X}^{\text{SC/SDP}}_{\text{IC}}$.

Table I compares key differences from various aspects among AD methods using LRaSR-based models along with their corresponding ADs.

VI. 3-D ROC CURVE-DERIVED DETECTION MEASURES

The 2-D ROC curve has widely been used to measure the performance analysis of AD. It is a plot of detection probability P_{D} versus false alarm probability P_{F}. Its AUC, denoted by $\text{AUC}_{(\text{D,F})}$, is calculated as an indicator to measure how effective a detector is. Unfortunately, according to a recent work in [56], solely relying on $\text{AUC}_{(\text{D,F})}$ is not creditable. This is because both P_{D} and P_{F} are calculated by the same threshold τ used by the detector. As a result, when both P_{D}

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and P_F are very high, its calculated $AUC_{(D,F)}$ is also very high. Similarly, when both P_D and P_F are very low, its calculated $AUC_{(D,F)}$ is also very low. In other words, P_D and P_F are tied together simultaneously and cannot work independently. Consequently, P_D and P_F cannot individually measure target detectability (TD) and BKG suppressibility (BS), respectively. Addressing this issue [56] developed an effective 3-D ROC analysis-based evaluation tool, which extends the traditional 2-D ROC analysis by including the threshold τ as an additional parameter to represent a 3-D RIOC curve as a function of three parameters, P_D, P_F, and τ. As a result, a 3-D ROC curve can be generated by a triplet parameter vector specified by (P_D,P_F,τ) and it can be also used to three 2-D ROC curves of (P_D,P_F), (P_D,τ), and (P_F,τ) with their respective AUC values, denoted by $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, and $AUC_{(F,\tau)}$, where $AUC_{(D,\tau)}$ and $AUC_{(F,\tau)}$ can be used to evaluate TD and BS, respectively. In addition to these three AUC values, five new AUC measures are also developed in [3] and [55] to measure joint TD, joint BS, target detection and BKG suppression (TDBS), signal-to-noise probability ratio (SNPR), and ODP defined in the following.

1) $AUC_{(D,F)}$: effectiveness of a detector

$$0 \le AUC_{(D,F)} \le 1. \tag{35}$$

2) $AUC_{(D,\tau)}$: TD of a detector

$$0 \le AUC_{(D,\tau))} \le 1. \tag{36}$$

3) $AUC_{(F,\tau)}$: BS of a detector

$$0 \le AUC_{(F,\tau)}) \le 1. \tag{37}$$

4) AUC_{TD}: joint TD of a detector defined by

$$0 \le AUC_{TD} = AUC_{(D,F)} + AUC_{(D,\tau)} \le 2. \tag{38}$$

5) AUC_{BS}: joint BS of detector defined by

$$-1 \le AUC_{BS} = AUC_{(D,F)} - AUC_{(F,\tau)} \le 1. \tag{39}$$

6) AUC_{TDBS}: TDBS of a detector defined by

$$-1 \le AUC_{TDBS} = AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 1. \tag{40}$$

7) AUC_{SNPR}: SNPR of a detector defined by

$$0 \le AUC_{SNPR} = \frac{AUC_{(D,\tau)}}{AUC_{(F,\tau)}}. \tag{41}$$

8) AUC_{ODP}: ODP of a detector defined by

$$-1 \le AUC_{ODP} = AUC_{(D,F)}+AUC_{(D,\tau)} - AUC_{(F,\tau)} \le 2. \tag{42}$$

To summarize, the above detection measures can be generally categorized into the following three functionalities.

1) TD: $AUC_{(D,\tau)}$, AUC_{TD};

2) BKG suppressibility (BS): $AUC_{(F,\tau)}$, AUC_{BS}, AUC_{SNPR}; and

3) detector's effectiveness: $AUC_{(D,F)}$, AUC_{TDBS}, and AUC_{ODP};

For those who are interested in using the 3-D ROC analysis developed in [56] for detection performance evaluation, a software package of calculating the above detection measures is available on the website of the Remote Sensing Signal and Image Processing Laboratory (RSSIPL) at the University of Maryland, Baltimore County (UMBC), https://wiki.umbc.edu/display/rssipl/10. +Download.

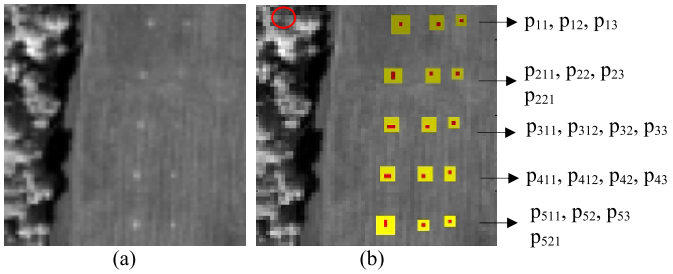


Fig. 4. (a) HYDICE 15-panel scene contains 15 panels. (b) Ground-truth map of spatial locations of 19 R panel pixels.

VII. Hyperspectral Image Experiments and Results

Three hyperspectral real image scenes were used for experiments.

A. Image Datasets to Be Used for Experiments

1) Hyperspectral Digital Imagery Collection Experiment (HYDICE) 15-Panel Scene: An airborne HYDICE scene shown in Fig. 4 was collected in August 1995 from a flight altitude of 10 000 ft with the ground sampling distance approximately 1.56 m. This scene has been studied extensively by many reports such as [1] and [3]. A total of 169 bands were used for the experiments with low signal/high noise bands: bands 1–3, and bands 202–210; and water vapor absorption bands: bands 101–112 and bands 137–153 removed. There are 15 square panels in Fig. 4(a) with three different sizes, 3 m × 3 m, 2 m × 2 m, and 1 m × 1 m, respectively. Due to the ground sampling distance approximately 1.56 m, each of the panels in the first column except for the first row contains two panel pixels highlighted by red, p_{211} and p_{221} in row 2, p_{311} and p_{312} in row 3, p_{411} and p_{412} in row 4, and p_{511} and p_{521} in row 5, as shown in Fig. 4. All the remaining 11 panels in Fig. 4 containing one single panel pixel for each panel also highlighted by red, p_{11}, p_{12}, and p_{13} in row 1, p_{22} and p_{23} in row 2, p_{32} and p_{33} in row 3, p_{42} and p_{43} in row 4, and p_{52} and p_{53} in row 5. Therefore, there are a total of 19 red (R) panel pixels. Fig. 4(b) shows their precise spatial locations with the pixels in yellow (Y pixels), indicating panel pixels mixed with the BKG. It should also be noted that at the left upper corner, there is a two-pixel size rock highlighted by a red open circle in Fig. 4(b), which can be considered as anomalous pixels not a target as panel pixels.

2) HYDICE Urban Scene: A second dataset is another HYDICE image with pseudo-color shown in Fig. 5(a). It is an urban scene and comprised 210 spectral bands with 174 bands being used for experiments after the noise and water absorption bands had been removed. A region with a size of 80 × 100 pixels located at the upper right of the scene was selected as the test image shown in Fig. 5(b) along with the ground-truth map shown in Fig. 5(c) where 21 pixels were

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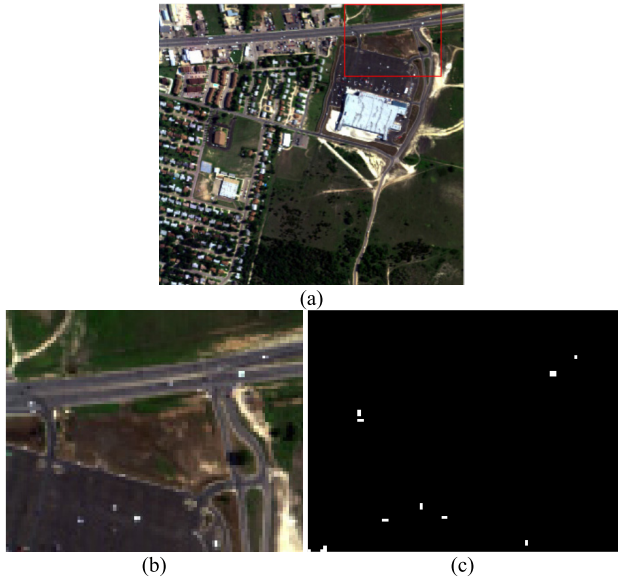


Fig. 5. HYDICE urban scene. (a) Pseudo-color image of the whole scene. (b) Pseudo-color image of the selected area. (c) Ground-truth map.

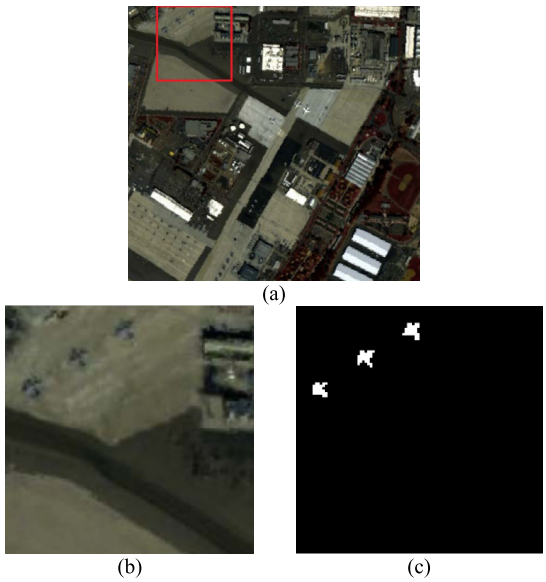


Fig. 6. San Diego airport scene. (a) Pseudo-color image of the whole scene. (b) Pseudo-color image of the selected area. (c) Ground-truth map.

identified as anomalies, which were cars and roofs because they had spectra that differ from BKG.

3) San Diego Airport Data: A third dataset used for experiments was also an airborne visible/infrared imaging spectrometer (AVIRIS) acquired scene, which is a San Diego airport scene in CA, USA, with pseudo-color shown in Fig. 6(a).

It has a size of 400 × 400 pixels with a 3.5-m spatial resolution and 224 spectral channels in wavelengths ranging from 370 to 2510 nm. After removing the bands that correspond to the water absorption regions, low signal-to-noise ratio, and bad bands (1–6, 33–35, 97, 107–113, 153–166, and 221–224), 189 available bands of the data were retained in the experiments. An area of 100 × 100 pixels at the upper left corner of the scene was selected as the test image shown in Fig. 6(b) along with its ground-truth map shown in Fig. 6(c). It is an urban scene in which the main BKG materials are roof, shadow, and grass. There are three airplanes in the image, which consist of 58 pixels and account for 0.33% of the test image, as shown in Fig. 6(b). This scene is less interesting than the HYDICE 15-panel data in Fig. 4, because only ground truth is available is three airplanes which are large targets.

TABLE II
p, m, j ESTIMATED FOR HYDICE DATA BY NWHFC USING MX-SVD

	p	*m*	*j*
HYDICE 15-panel scene	13	7	6
HYDICE urban scene	17	11	6
San Diego airport	11	2	9

Table II calculated the value of p estimated by noise whiten HFC (NWHFC) methods and the value of m estimated by MX-SVD for all the three scenes.

B. Experimental Results and Discussions

In order to evaluate the performance of IC-SDP-AD, three key features were for comparison. One is ADs to be used, which are RX-AD [4] and OSP-AD [52]. Since RX-AD and CEM-AD [11] perform very closely and similarly, only RX-AD was used for experiments. Another is BKG removal, which is DS, GoDec [43], and CDA [44]. A third one is anomaly space in which an AD operates. It imposes SC introduced by GoDec and SDP developed in Section III on the sphered data space. Therefore, many combinations can be derived from these three features. In the following experiments, we have particularly focused on the anomaly space obtained by the IC component space coupled with SC and SDP. In particular, since OSP-AD requires $\mathbf{U}$ for annihilation, GoDec was further implemented in the DS-space to find its low rank space to be used as $\mathbf{U}$. As a consequence, a total of eight scenarios for each of RX-AD and OSP-AD were performed for experiments where DS-IC-SC/SDP is abbreviated as ICSC/IC-SDP.

1) Reed-Xiaoli Anomaly Detector (RX-AD):

$$\delta^{\text{RX-AD}}_{\text{ICSC}},\quad \delta^{\text{RX-AD}}_{\text{IC-}j\text{-FSDP}},\quad \delta^{\text{RX-AD}}_{\text{IC-}j\text{-BSDP}},\quad \delta^{\text{RX-AD}}_{\text{IC-SSDP}}$$
$$\delta^{\text{RX-AD}}_{\text{CDASC}},\quad \delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FSDP}},\quad \delta^{\text{RX-AD}}_{\text{CDA-}j\text{-BSDP}},\quad \delta^{\text{RX-AD}}_{\text{CDA-SSDP}}$$

2) Orthogonal Subspace Projection Anomaly Detector (OSP-AD):

$$\delta^{\text{OSP-AD}}_{\text{ICSC}},\quad \delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FSDP}},\quad \delta^{\text{OSP-AD}}_{\text{IC-}j\text{-BSDP}},\quad \delta^{\text{OSP-AD}}_{\text{IC-SSDP}},$$
$$\delta^{\text{OSP-AD}}_{\text{CDASC}},\ \delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FSDP}},\quad \delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-BSDP}},\quad \delta^{\text{OSP-AD}}_{\text{CDA-SSDP}}.$$

It should be noted that many ADs including CRD [18], RPCA [34], and LRaSMD-based Mahalanobis distance AD (LSMAD) in [56] were compared in [51], and their detailed comparative analyses can be found in [44], [51], and [52] where CDASC in [44] has shown to perform better than most recently developed AD methods. In this case, we only focus on comparative studies between IC-SC/SDP-AD and CDASC.

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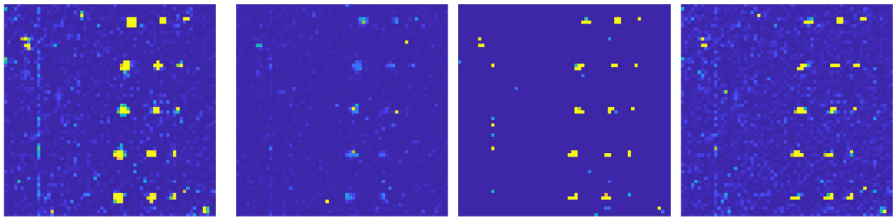


Fig. 7. Detection maps of HYDICE 15-panel scene using RX-AD on IC using SC and SDP with $p = 13$, $m = 7$, and $j = 6$. (a) $\delta^{\text{RXAD}}_{\text{ICSC}}$. (b) $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FS-SDP}}$. (c) $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FB-SDP}}$. (d) $\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$.

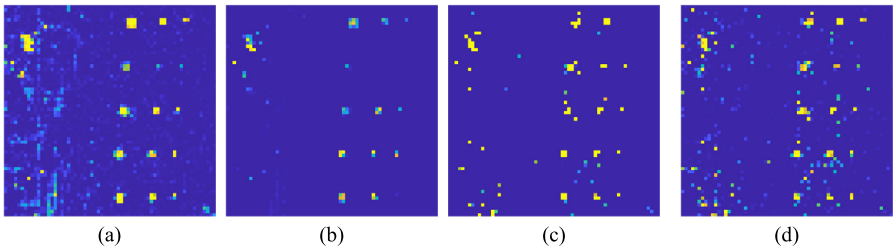


Fig. 8. Detection maps of HYDICE 15-panel scene using RX-AD on CDA using SC and SDP with $p = 13$, $m = 7$, and $j = 6$. (a) $\delta^{\text{RX-AD}}_{\text{CDASC}}$. (b) $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FS-SDP}}$. (c) $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FB-SDP}}$. (d) $\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$.

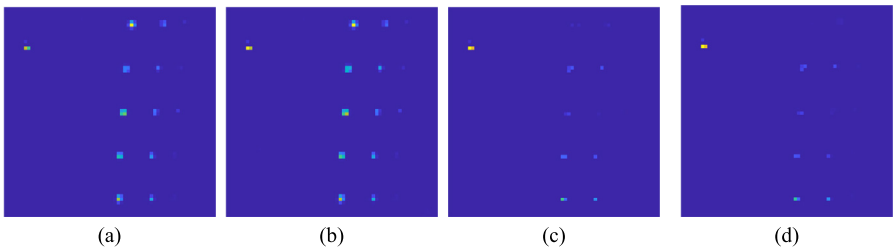


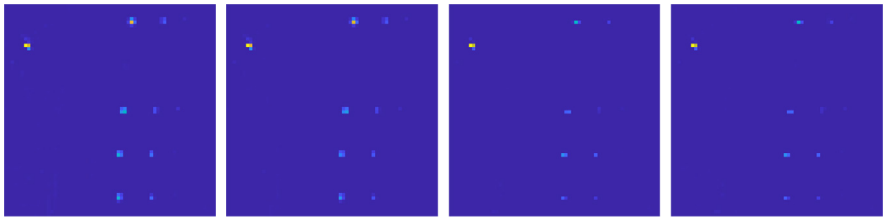
Fig. 9. Detection maps of HYDICE 15-panel scene using OSP-AD on IC using SC and SDP with $p = 13$, $m = 7$, and $j = 6$. (a) $\delta^{\text{OSP-AD}}_{\text{ICSC}}$. (b) $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$. (c) $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FB-SDP}}$. (d) $\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$.

VIII. HYDICE 15-Panel Scene

Fig. 7 shows the detection maps of RX-AD implemented on IC component space using SC and SDP, while Fig. 8 shows the detection maps of RX-AD implemented in the CDA component space using SC and SDP. Comparing Figs. 7 and 8 by visual inspection, it seems that RX-AD using IC-SC/SDP performed better than RX-AD using CDA-SC/SDP in terms of anomaly detectability and BKG suppression. Specifically, RX-AD using CDA-specified sparse space produced more false alarm pixels than IC-specified sparse space does. Similarly, Figs. 9 and 10 show the detection maps of OSP-AD implemented on IC component space using SC and SDP and CDA component space using SC and SDP, respectively. It is very clear to see by comparing the results in Figs. 7–10 that OSP-AD achieved significantly better than RX-AD at the expense of anomaly detectability where the BKG was virtually wiped out by OSP rather than suppressed by inverting the covariance matrix $\mathbf{K}$ by RX-AD. It is worth noting that from the ground truth of Fig. 4(b), the two-pixel size anomaly was clearly detected by OSP-AD in Figs. 9 and 10 not by RX-AD in Figs. 7 and 8. These results demonstrated how crucial BKG played a key role in assessing the effectiveness of an AD.

It is apparent from Figs. 7 to 10 to show that the traditional 2-D ROC curve analysis alone could not be effective in evaluating BKG. To illustrate this crucial impact, Tables III and III conducted a 3-D ROC analysis among the detection results in Figs. 7–10 using eight detection measures presented in Section V where the best results are boldfaced. Generally speaking, RX-AD and OSP-AD using ICSC in Table III were more effective than RX-AD and OSP-AD using CDASC in Table IV. In addition, if we looked at $\text{AUC}_{(\text{D,F})}$ alone, OSP-AD did the best. On the whole, when RX-AD was used, ICSC was the best, i.e., $\delta^{\text{RX-AD}}_{\text{ICSC}}$ was the best. In contrast, when OSP-AD was used, the IC-j-FS-SDP was the best, that is, $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$ was the best. The conclusions can be summarized as follows. If we are interested in TD, $\delta^{\text{RX-AD}}_{\text{ICSC}}$ is the best in detecting all the five subpixel panel pixels in the third column at the expense of many falsely alarmed pixels. In contrast, if we are interested in BKG suppression, $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$ was the best to detect all the 14-panel pixels in the first two columns plus the two-pixel anomaly highlighted by red with missing detection of subpixel panels in the third column. The results in Tables III and IV suggested that SDP worked better than SC in dealing with sparse signals. In general, it is also clearly shown that OSP-AD

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(a) (b) (c) (d)

Fig. 10. Detection maps of HYDICE 15-panel scene using OSP-AD on CDA using SC and SDP with $p = 13$, $m = 7$, and $j = 6$. (a) $\delta^{\text{OSP-AD}}_{\text{CDASC}}$. (b) $\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FB-SDP}}$. (c) $\delta^{\text{OSP-AD}}_{\text{CDA-}J\text{-FB-SDP}}$. (d) $\delta^{\text{OSP-AD}}_{\text{CDA-SS-SDP}}$.

TABLE III
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FROM FIGS. 7 TO 8 PRODUCED BY RX-AD FOR HYDICE PANEL DATA

detector	$\text{AUC}_{(\text{D,F})}$	$\text{AUC}_{(\text{D},\tau)}$	$\text{AUC}_{(\text{F},\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{RX-AD}}_{\text{ICSC}}$	**0.9958**	**0.9965**	0.0370	**1.9922**	0.9585	26.9263	**0.9595**	**1.9552**	**0.157578 s**
$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FS-SDP}}$	0.9639	0.2337	0.0129	1.1976	0.9508	18.1068	0.2208	1.1846	1.716624 s
$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FB-SDP}}$	0.9706	0.9223	0.0913	1.8929	0.8791	10.1041	0.8310	1.8016	0.243244 s
$\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$	0.9793	0.9487	0.0343	1.9280	0.9447	27.6191	0.9143	1.8936	0.244005 s
$\delta^{\text{RX-AD}}_{\text{CDASC}}$	0.9936	0.8949	0.0357	1.8885	0.9575	25.0649	0.8592	1.8528	0.201388 s
$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FS-SDP}}$	0.9398	0.4519	**0.0106**	1.3917	0.9290	**42.7047**	0.4413	1.3811	1.872864 s
$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FB-SDP}}$	0.9674	0.9478	0.0228	1.9152	0.9444	41.6306	0.9250	1.8924	0.536954 s
$\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$	0.9936	0.9404	0.0257	1.9340	**0.9677**	36.6605	0.9148	1.9083	0.305117 s

TABLE IV
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FROM FIGS. 9 TO 10 PRODUCED BY OSP-AD FOR HYDICE PANEL DATA

detector	$\text{AUC}_{(\text{D,F})}$	$\text{AUC}_{(\text{D},\tau)}$	$\text{AUC}_{(\text{F},\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{OSP-AD}}_{\text{ICSC}}$	**0.9974**	0.3530	0.0018	1.3503	**0.9953**	195.0958	0.3512	1.3485	**0.107851 s**
$\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$	0.9974	**0.4042**	0.0023	**1.4016**	0.9950	178.1884	**0.4019**	**1.3993**	1.505915 s
$\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FB-SDP}}$	0.9718	0.2264	0.0007	1.1982	0.9709	337.8551	0.2257	1.1975	0.217818 s
$\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$	0.9591	0.2342	0.0010	1.1934	0.9579	**237.9221**	0.2333	1.1924	0.183861 s
$\delta^{\text{OSP-AD}}_{\text{CDASC}}$	0.9912	0.2126	0.0019	1.2038	0.9891	114.0441	0.2107	1.2019	0.146984 s
$\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FS-SDP}}$	0.9405	0.1689	0.0020	1.1094	0.9383	85.4518	0.1669	1.1075	1.575384 s
$\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FB-SDP}}$	0.9697	0.1168	**0.0006**	1.0865	0.9688	188.6090	0.1162	1.0859	0.308981 s
$\delta^{\text{OSP-AD}}_{\text{CDA-SS-SDP}}$	0.9910	0.1208	0.0008	1.1118	0.9900	157.2892	0.1201	1.1111	0.222997 s

performed BKG annihilation by OSP which was much better than BKG suppression performed by RX-AD using $\mathbf{K}^{-1}$.

IX. HYDICE URBAN SCENE

Unlike HYDICE panel data scene which provides the detailed panel pixel information for image analysis, this HYDICE urban scene is less interesting because the provided information about anomalous target pixels is limited. Fig. 11 shows the detection maps of RX-AD implemented on IC component space using SC and SDP, while Fig. 12 shows the detection maps of RX-AD implemented on CDA component space using SC and SDP. Similarly, Fig. 13 shows the detection maps of OSP-AD implemented on IC component space using SC and SDP, while Fig. 14 shows the detection maps of OSP-AD implemented on CDA component space using SC and SDP. By visual inspection of Figs. 11–14, it is clear to show that OSP-AD performed significantly better than RX-AD on the IC component space in BKG suppression. In particular, when RX-AD was implemented, j-FS-SDP performed better than SC, j-FB-SDP, and SS-SDP on both IC and CDA.

Obviously, assessment by visual inspection may not be reliable. Tables V and VI conducted a quantitative comparison based on 3-D ROC analysis obtained from the detection results in Figs. 11–14 where the best results are boldfaced. On the whole, RX-AD and OSP-AD using ICSC performed the best. However, if BKG suppression is of major concern, RX-AD and OSP-AD using j-FS-SDP were the best.

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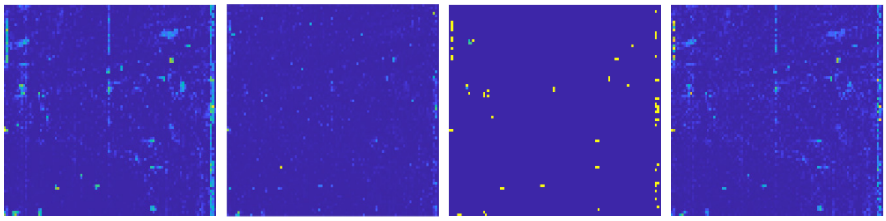


Fig. 11. Detection maps of HYDICE URBAN scene using RX-AD on IC using SC and SDP with $p = 17$, $m = 11$, and $j = 6$. (a) $\delta^{\text{RXAD}}_{\text{ICSC}}$. (b) $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FS-SDP}}$. (c) $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FB-SDP}}$. (d) $\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$.

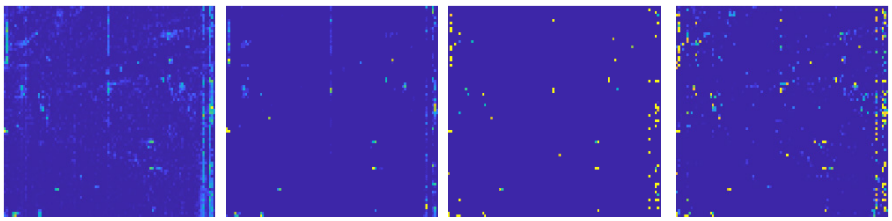


Fig. 12. Detection maps of HYDICE URBAN scene using RX-AD on CDA using SC and SDP with $p = 17$, $m = 11$, and $j = 6$. (a) $\delta^{\text{RXAD}}_{\text{CDASC}}$. (b) $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FS-SDP}}$. (c) $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FB-SDP}}$. (d) $\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$.

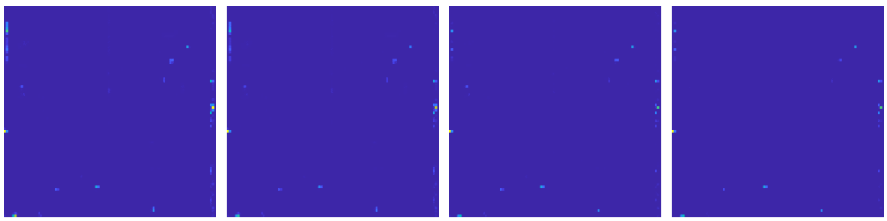


Fig. 13. Detection maps of HYDICE URBAN scene using OSP-AD on IC using SC and SDP with $p = 17$, $m = 11$, and $j = 6$. (a) $\delta^{\text{OSP-AD}}_{\text{ICSC}}$. (b) $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$. (c) $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FB-SDP}}$. (d) $\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$.

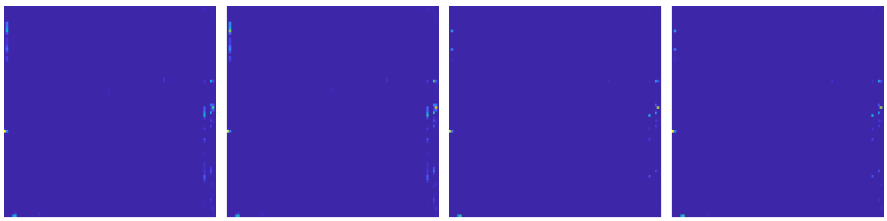


Fig. 14. Detection maps of HYDICE URBAN scene using OSP-AD on CDA using SC and SDP with $p = 17$, $m = 11$, and $j = 6$. (a) $\delta^{\text{OSP-AD}}_{\text{CDASC}}$. (b) $\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FS-SDP}}$. (c) $\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FB-SDP}}$. (d) $\delta^{\text{OSP-AD}}_{\text{CDA-SS-SDP}}$.

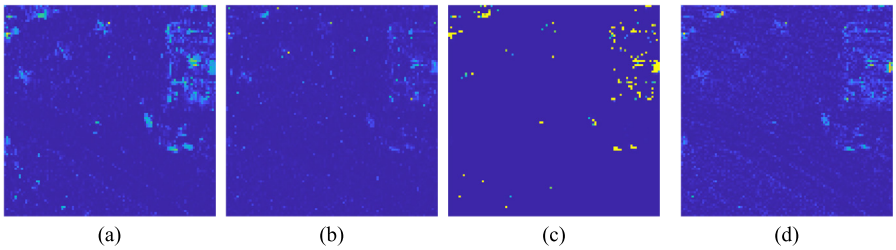


Fig. 15. Detection maps of HYDICE San Diego scene using RX-AD on IC using SC and SDP with $p = 11$, $m = 2$, and $j = 9$. (a) $\delta^{\text{RXAD}}_{\text{ICSC}}$. (b) $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FS-SDP}}$. (c) $\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FB-SDP}}$. (d) $\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$.

X. San Diego Airport Scene

This scene is interesting because the major targets of interest are three airplanes. Technically speaking, these airplanes cannot be considered as anomalies because they are so large and visible. This fact was also demonstrated in [44] where the airplanes could only be detected when the BKG was included

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TABLE V

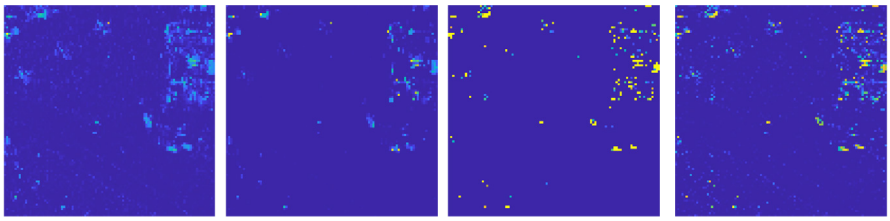
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FROM FIGS. 11 TO 12 PRODUCED BY RX-AD FOR HYDICE URBAN DATA

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{RX-AD}}_{\text{ICSC}}$	**0.9956**	0.5784	0.0233	**1.5740**	0.9722	24.8078	**0.5551**	**1.5507**	**0.157578 s**
$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FS-SDP}}$	0.9820	0.1947	**0.0095**	1.1767	**0.9724**	20.4160	0.1852	1.1672	1.716624 s
$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FB-SDP}}$	0.8057	**0.6514**	0.1646	1.4571	0.6411	3.9584	0.4868	1.2926	0.243244 s
$\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$	0.9866	0.4113	0.0223	1.3979	0.9641	18.4046	0.3890	1.3755	0.244005 s
$\delta^{\text{RX-AD}}_{\text{CDASC}}$	0.9796	0.3522	0.0229	1.3319	0.9566	15.3978	0.3294	1.3090	0.255149 s
$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FS-SDP}}$	0.7981	0.2052	0.0120	1.0032	0.7860	17.1503	0.1932	0.9913	4.913696 s
$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FB-SDP}}$	0.7328	0.4049	0.0259	1.1377	0.7068	15.6032	0.3789	1.1118	0.584837 s
$\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$	0.9288	0.5027	0.0192	1.4315	0.9094	**26.1756**	0.4835	1.4123	0.469419 s

TABLE VI

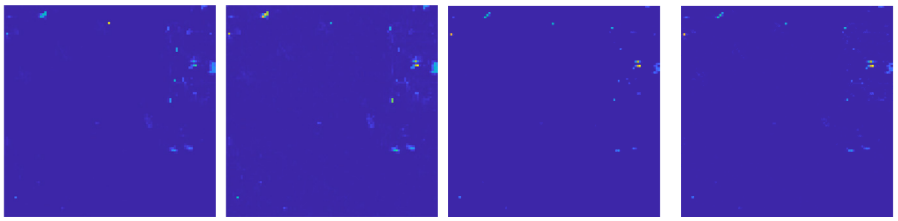
AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FROM FIGS. 13 TO 14 PRODUCED BY OSP-AD FOR HYDICE URBAN DATA

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{OSP-AD}}_{\text{ICSC}}$	**0.9955**	**0.1930**	0.0016	**1.1885**	**0.9937**	118.6888	**0.1914**	**1.1869**	**0.107851 s**
$\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$	0.9954	0.1441	0.0018	1.1395	0.9935	81.8230	0.1423	1.1377	1.505915 s
$\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$	0.9815	0.1468	0.0009	1.1283	0.9805	164.8928	0.1459	1.1274	0.183861 s
$\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FB-SDP}}$	0.8071	0.1185	**0.0005**	0.9256	0.8065	**221.5314**	0.1180	0.9251	0.217818 s
$\delta^{\text{OSP-AD}}_{\text{CDASC}}$	0.9759	0.0513	0.0016	1.0272	0.9742	31.7904	0.0496	1.0256	0.230808 s
$\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FS-SDP}}$	0.7947	0.0501	0.0023	0.8448	0.7923	22.2696	0.0479	0.8425	5.003275 s
$\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FB-SDP}}$	0.7335	0.0474	0.0007	0.7809	0.7328	72.6895	0.0467	0.7802	0.563364 s
$\delta^{\text{OSP-AD}}_{\text{CDA-SS-SDP}}$	0.9329	0.0491	0.0008	0.9820	0.9320	64.8117	0.0484	0.9812	0.384178 s



(a) (b) (c) (d)

Fig. 16. Detection maps of HYDICE San Diego scene using RX-AD on CDA using SC and SDP with $p = 11$, $m = 2$, and $j = 9$. (a) $\delta^{\text{RX-AD}}_{\text{CDASC}}$. (b) $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FS-SDP}}$. (c) $\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FB-SDP}}$. (d) $\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$.

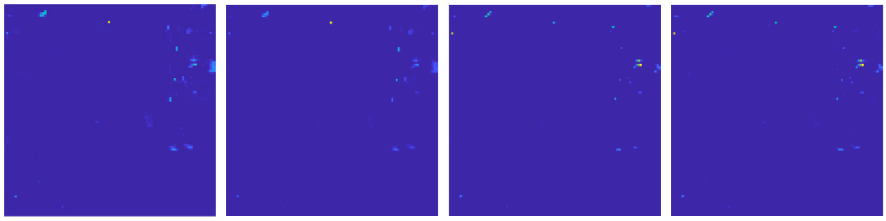


(a) (b) (c) (d)

Fig. 17. Detection maps of HYDICE San Diego scene using OSP-AD on IC using SC and SDP with $p = 11$, $m = 2$, and $j = 9$. (a) $\delta^{\text{OSP-AD}}_{\text{ICSC}}$. (b) $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$. (c) $\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FB-SDP}}$. (d) $\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$.

in RX-AD and OSP-AD. Therefore, when RX-AD and OSP-AD use $\mathbf{X}^{\text{SC}}_{\mathbf{IC}^j}$ or $\mathbf{X}^{\text{SDP}}_{\mathbf{IC}^j}$ as its data space, it is expected that these airplanes would not be detected. Figs. 15–18 showed exactly the case when both RX-AD and OSP-AD were implemented on IC component space using SC and SDP where many small targets other than airplanes were detected. Unfortunately, the

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(a) (b) (c) (d)

Fig. 18. Detection maps of HYDICE San Diego scene using OSP-AD on CDA using SC and SDP with $p = 11$, $m = 2$, and $j = 9$. (a) $\delta^{\text{OSP-AD}}_{\text{CDASC}}$. (b) $\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FS-SDP}}$. (c) $\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FB-SDP}}$. (d) $\delta^{\text{OSP-AD}}_{\text{CDA-SS-SDP}}$.

TABLE VII

AUC VALUES CALCULATED FROM THREE 2 -D ROC CURVES FROM FIGS. 15 TO 16 PRODUCED BY RX-AD FOR SAN DIEGO AIRPORT DATA

detector	$\text{AUC}_{(D,F)}$	$\text{AUC}_{(D,\tau)}$	$\text{AUC}_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{RX-AD}}_{\text{ICSC}}$	**0.8530**	0.1158	0.0242	**0.9687**	0.8270	**4.7894**	**0.0916**	**0.9446**	0.387074 s
$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FS-SDP}}$	0.8383	0.0594	0.0130	0.8977	0.8233	4.5638	0.0464	0.8847	4.320170 s
$\delta^{\text{RX-AD}}_{\text{IC-}j\text{-FB-SDP}}$	0.8723	0.0615	0.0149	0.9338	0.8573	4.1188	0.0466	0.9189	0.751438 s
$\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$	0.7453	0.0709	0.0237	0.8162	0.7216	2.9944	0.0472	0.7925	0.614247 s
$\delta^{\text{RX-AD}}_{\text{CDASC}}$	0.8492	0.0863	0.0195	0.9355	**0.8296**	4.4150	0.0668	0.9160	**0.357907 s**
$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FS-SDP}}$	0.7565	0.0237	**0.0075**	0.7802	0.7490	3.1592	0.0162	0.7727	6.156887 s
$\delta^{\text{RX-AD}}_{\text{CDA-}j\text{-FB-SDP}}$	0.5552	**0.1316**	0.0641	0.6868	0.4910	2.0534	0.0675	0.6227	0.783935 s
$\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$	0.7225	0.0836	0.0181	0.8061	0.7043	4.6228	0.0655	0.7880	0.549688 s

TABLE VIII

AUC VALUES CALCULATED FROM THE THREE 2-D ROC CURVES FROM FIGS. 17 TO 18 PRODUCED BY OSP-AD FOR SAN DIEGO AIRPORT DATA

detector	$\text{AUC}_{(D,F)}$	$\text{AUC}_{(D,\tau)}$	$\text{AUC}_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{OSP-AD}}_{\text{ICSC}}$	**0.8769**	**0.0167**	0.0027	**0.8937**	**0.8728**	6.1958	**0.0140**	**0.8910**	0.309286 s
$\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FS-SDP}}$	0.8528	0.0150	0.0059	0.8678	0.8468	2.5376	0.0091	0.8619	3.845969 s
$\delta^{\text{OSP-AD}}_{\text{IC-}j\text{-FB-SDP}}$	0.5391	0.0057	0.0012	0.5448	0.5378	4.8576	0.0046	0.5436	0.643991 s
$\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$	0.7096	0.0106	0.0028	0.7203	0.7067	3.8081	0.0078	0.7175	0.492797 s
$\delta^{\text{OSP-AD}}_{\text{CDASC}}$	0.8616	0.0161	0.0027	0.8777	0.8572	5.9496	0.0134	0.8750	**0.295419 s**
$\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FS-SDP}}$	0.7589	0.0129	0.0015	0.7717	0.7545	**8.8005**	0.0114	0.7703	5.513146 s
$\delta^{\text{OSP-AD}}_{\text{CDA-}j\text{-FB-SDP}}$	0.5551	0.0055	**0.0011**	0.5606	0.5540	5.1266	0.0044	0.5595	0.790877 s
$\delta^{\text{OSP-AD}}_{\text{CDA-SS-SDP}}$	0.7247	0.0086	0.0016	0.7333	0.7230	5.2285	0.0069	0.7317	0.565342 s

lack of ground truth and most detected targets were unknown and difficult to verify and identify. Nevertheless, comparing Figs. 15 and 16 with Figs. 17 and 18 by visual inspection, OSP-AD did significantly better than RX-AD in the sense of BKG removal versus BKG suppression.

Despite that we do not consider the three airplanes as anomalies, using their provided ground truth, Tables VII and VIII tabulate the eight detection measures generated from their 3-D ROC curves obtained by RX-AD and OSP-AD in Figs. 15–18, respectively, where the best results are boldfaced. Generally speaking, $\delta^{\text{RX-AD}}_{\text{ICSC}}$ and $\delta^{\text{OSP-AD}}_{\text{ICSC}}$ seemed to perform better than all other ADs. Nevertheless, CDASC developed in [44] also performed comparably to ICSC. In addition, shown in Tables VII and VIII, OSP-AD achieved much better BKG removal than BKG suppression by RX-AD.

XI. COMPARATIVE ANALYSIS WITH LATEST DEVELOPED AD APPROACHES

To further demonstrate the superior performance of the proposed DS-SC/SDP-AD, this section compares several latest developed methods for AD where their algorithms are available on the website and can be used for comparative performance analysis. For example, the LRR-based method developed in [40] is to represent the category of LRaSR model-based methods. In addition, [27] represents the category of

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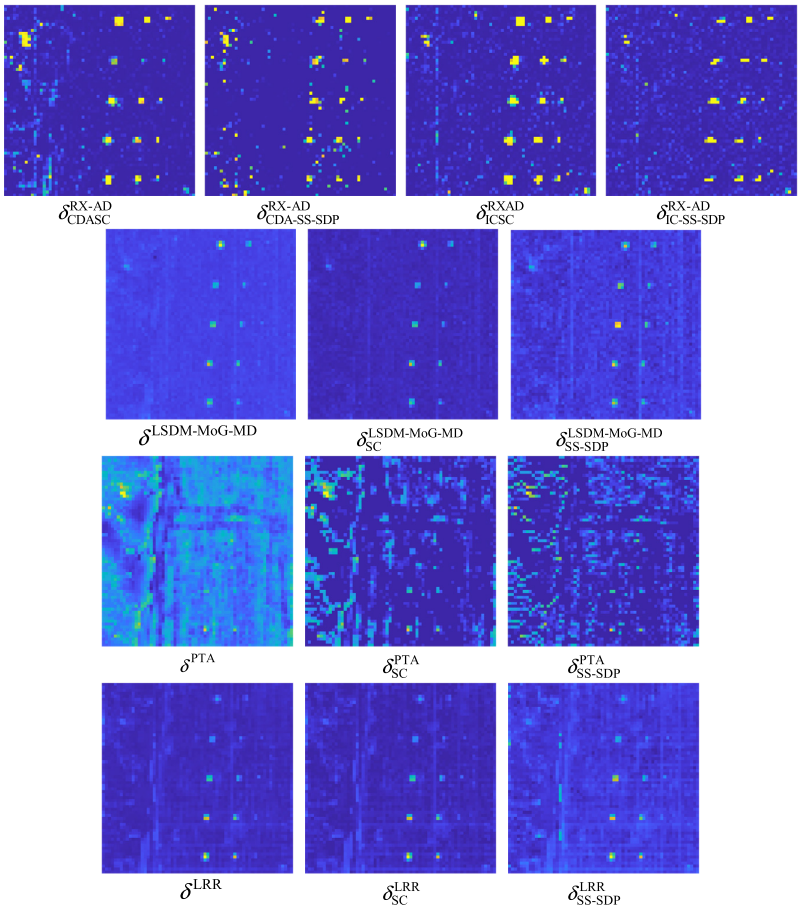


Fig. 19. AD maps produced by $\delta^{\text{RX-AD}}_{\text{CDASC}}$, $\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$, $\delta^{\text{RXAD}}_{\text{ICSC}}$, $\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$, $\delta^{\text{LSDM-MoG-MD}}$, $\delta^{\text{LSDM-MoG-MD}}_{\text{SC}}$, $\delta^{\text{LSDM-MoG-MD}}_{\text{SS-SDP}}$, δ^{PTA}, $\delta^{\text{PTA}}_{\text{SC}}$, $\delta^{\text{PTA}}_{\text{SS-SDP}}$, δ^{LRR}, $\delta^{\text{LRR}}_{\text{SC}}$, and $\delta^{\text{LRR}}_{\text{SS-SDP}}$ for the HYDICE panel data using CDASC, ICSC, LSDM-MoG, PTA, and LRR.

TABLE IX

AUC OF HYDICE 15-PANEL SCENE PRODUCED IN FIG. 22 BY CDASC, ICSC, LSDM-MOG, PTA, AND LRRS WITH p = 13 AND j = 6

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{RX-AD}}_{\text{CDASC}}$	0.9936	0.8949	0.0357	1.8885	0.9575	25.0649	0.8592	1.8528	0.201388 s
$\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$	**0.9936**	**0.9404**	**0.0257**	**1.9340**	**0.9677**	**36.6605**	**0.9148**	**1.9083**	0.305117 s
$\delta^{\text{RX-AD}}_{\text{ICSC}}$	**0.9958**	**0.9965**	0.0370	**1.9922**	0.9585	26.9263	**0.9595**	**1.9552**	**0.157578 s**
$\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$	0.9793	0.9487	**0.0343**	1.9280	0.9447	**27.6191**	0.9143	1.8936	0.244005 s
$\delta^{\text{LSDM-MoG-MD}}$	0.9933	0.4400	0.0817	1.4332	0.9112	5.3884	0.3583	1.3516	0.603066 s
$\delta^{\text{LSDM-MoG-MD}}_{\text{SC}}$	0.9936	0.4016	**0.0306**	1.3952	**0.9627**	**13.1327**	0.3710	1.3646	0.832943 s
$\delta^{\text{LSDM-MoG-MD}}_{\text{SS-SDP}}$	**0.9963**	**0.5671**	0.0852	**1.5634**	0.9110	6.6589	**0.4819**	**1.4782**	0.804012 s
δ^{PTA}	0.5953	**0.3739**	0.2799	0.9692	0.3153	1.3360	0.0940	0.6894	0.080377 s
$\delta^{\text{PTA}}_{\text{SC}}$	0.7042	0.2704	0.0841	0.9747	0.6199	3.2155	**0.1863**	0.8906	0.116374 s
$\delta^{\text{PTA}}_{\text{SS-SDP}}$	**0.7153**	0.2627	**0.0777**	**0.9779**	**0.6374**	**3.3797**	**0.1849**	**0.9002**	0.184819 s
δ^{LRR}	0.9696	0.4186	0.0387	1.3882	0.9292	10.8266	0.3800	1.3495	0.111180 s
$\delta^{\text{LRR}}_{\text{SC}}$	0.9682	0.4247	**0.0351**	1.3929	**0.9314**	**12.1043**	0.3896	1.3578	0.150837 s
$\delta^{\text{LRR}}_{\text{SS-SDP}}$	**0.9808**	**0.4915**	0.0885	**1.4723**	0.8912	5.5520	**0.4030**	**1.3838**	0.237264 s

modeling BKG as Gaussian distributions and is a LSDM-MoG which models the sparse component by a mixture of Gaussian noises so that anomalies can be detected by the Manhattan distance. In the category of BKG-anomaly decomposition, Li *et al.* [42] developed that a PTA was developed to decompose the data space into a BKG tensor and an anomaly

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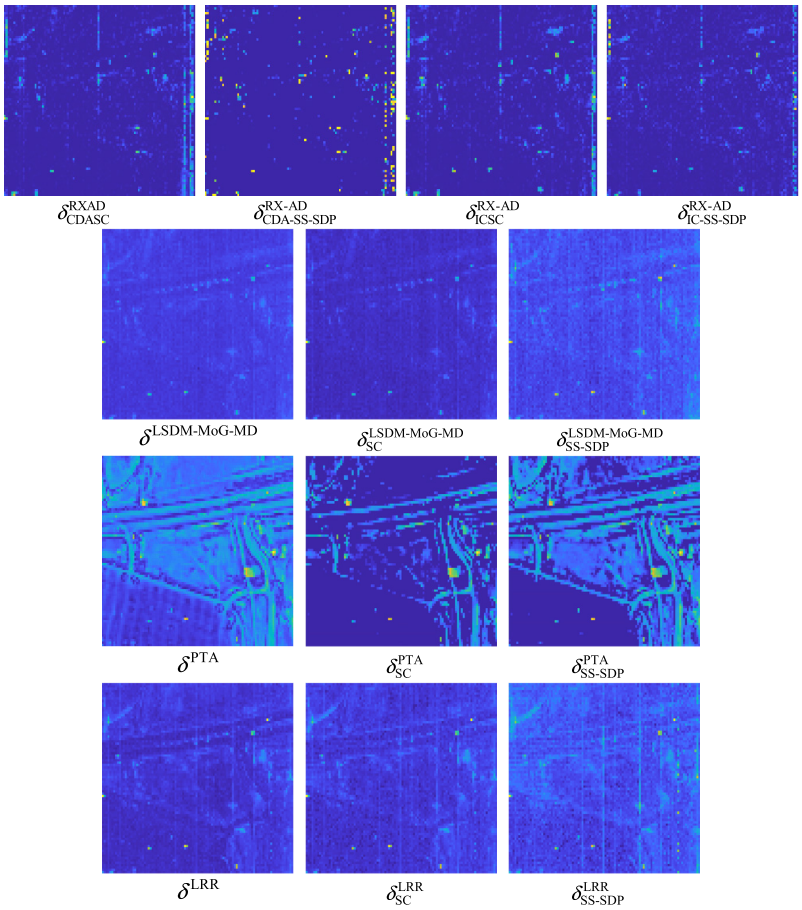


Fig. 20. AD maps produced by $\delta^{\text{RXAD}}_{\text{CDASC}}$, $\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$, $\delta^{\text{RX-AD}}_{\text{ICSC}}$, $\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$, $\delta^{\text{LSDM-MoG-MD}}$, $\delta^{\text{LSDM-MoG-MD}}_{\text{SC}}$, $\delta^{\text{LSDM-MoG-MD}}_{\text{SS-SDP}}$, δ^{PTA}, $\delta^{\text{PTA}}_{\text{SC}}$, $\delta^{\text{PTA}}_{\text{SS-SDP}}$, δ^{LRR}, $\delta^{\text{LRR}}_{\text{SC}}$, and $\delta^{\text{LRR}}_{\text{SS-SDP}}$ for the HYDICE urban data using CDASC, ICSC, LSDM-MoG, PTA, and LRR.

TABLE X

AUC OF HYDICE URBAN SCENE PRODUCED IN FIG. 23 BY CDASC, ICSC, LSDM-MOG, PTA, AND LRR WITH p = 17 AND j = 6

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{RX-AD}}_{\text{CDASC}}$	**0.9796**	0.3522	0.0229	1.3319	**0.9566**	15.3978	0.3294	1.3090	**0.255149 s**
$\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$	0.9288	**0.5027**	**0.0192**	**1.4315**	0.9094	**26.1756**	**0.4835**	**1.4123**	0.469419 s
$\delta^{\text{RX-AD}}_{\text{ICSC}}$	**0.9956**	**0.5784**	0.0233	**1.5740**	**0.9722**	**24.8078**	**0.5551**	**1.5507**	**0.157578 s**
$\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$	0.9866	0.4113	0.0223	1.3979	0.9641	18.4046	0.3890	1.3755	0.244005 s
$\delta^{\text{LSDM-MoG-MD}}$	**0.9834**	0.3596	0.0919	1.3430	0.8914	3.9116	0.2676	1.2511	0.767513 s
$\delta^{\text{LSDM-MoG-MD}}_{\text{SC}}$	0.9702	0.3033	**0.0480**	1.2735	**0.9221**	**6.3135**	0.2552	1.2255	0.856606 s
$\delta^{\text{LSDM-MoG-MD}}_{\text{SS-SDP}}$	0.9737	**0.5307**	0.1438	**1.5044**	0.8298	3.6902	**0.3869**	**1.3606**	1.046175 s
δ^{PTA}	0.7990	0.4457	0.2265	1.2447	0.5724	1.9676	0.2192	1.0182	0.079941 s
$\delta^{\text{PTA}}_{\text{SC}}$	**0.8379**	0.3771	**0.0742**	1.2150	**0.7636**	**5.0831**	**0.3029**	**1.1408**	0.139765 s
$\delta^{\text{PTA}}_{\text{SS-SDP}}$	0.8246	**0.4656**	0.1670	**1.2901**	0.6575	2.7879	0.2986	1.1231	0.284774 s
δ^{LRR}	**0.9937**	0.4965	**0.0697**	1.4902	**0.9239**	**7.1221**	**0.4268**	**1.4205**	0.131199 s
$\delta^{\text{LRR}}_{\text{SC}}$	0.9916	0.5104	0.0878	1.5020	0.9037	5.8138	0.4226	1.4142	0.181308 s
$\delta^{\text{LRR}}_{\text{SS-SDP}}$	0.9898	**0.5768**	0.1624	**1.5666**	0.8269	3.5508	0.4144	1.4042	0.455209 s

tensor for hyperspectral AD so that the anomalies can be detected in the anomaly tensor. Most recently, Chen *et al.* [44] derived a CDA which models data space as a three-component decomposition, principal component, IC, and noise component spaces. All these four approaches were used to conduct a comparative analysis.

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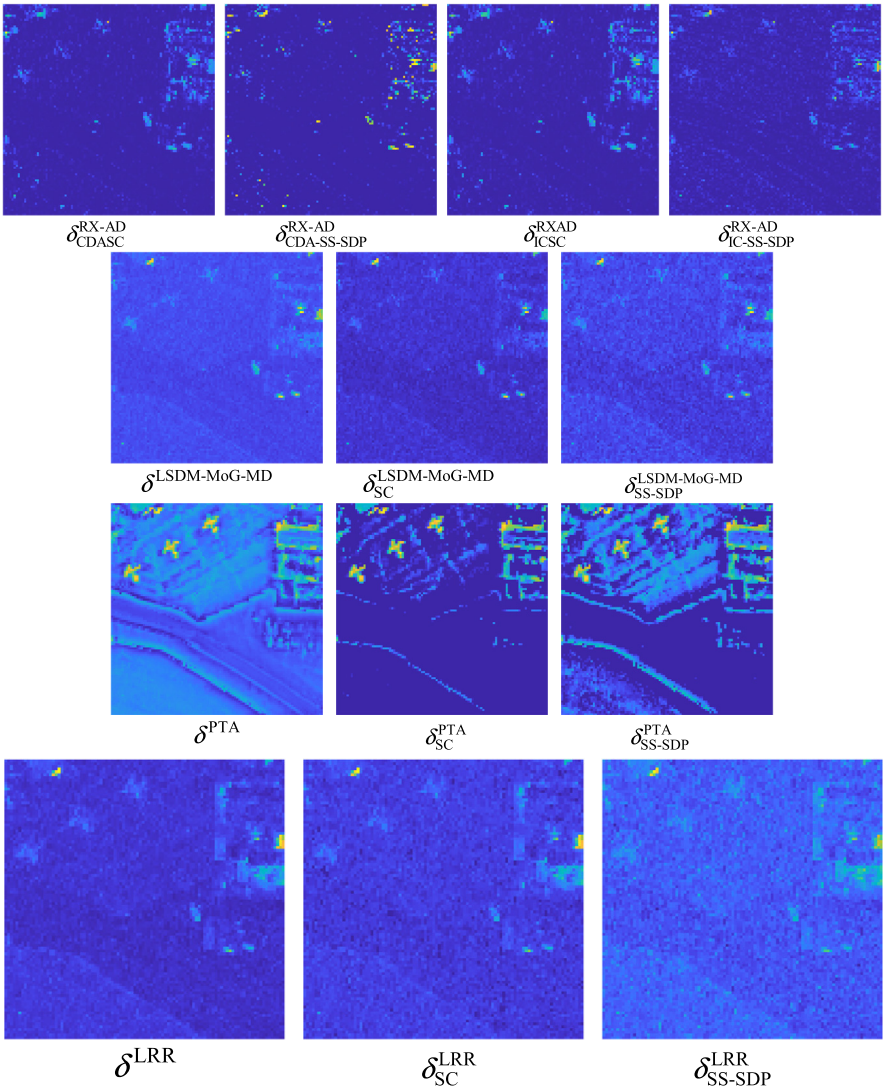


Fig. 21. AD maps produced by $\delta^{\text{RX-AD}}_{\text{CDASC}}$, $\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$, $\delta^{\text{RXAD}}_{\text{ICSC}}$, $\delta^{\text{RX-AD}}_{\text{IC-SS-SDP}}$, $\delta^{\text{LSDM-MoG-MD}}$, $\delta^{\text{LSDM-MoG-MD}}_{\text{SC}}$, $\delta^{\text{LSDM-MoG-MD}}_{\text{SS-SDP}}$, δ^{PTA}, $\delta^{\text{PTA}}_{\text{SC}}$, $\delta^{\text{PTA}}_{\text{SS-SDP}}$, δ^{LRR}, $\delta^{\text{LRR}}_{\text{SC}}$, and $\delta^{\text{LRR}}_{\text{SS-SDP}}$ for the San Diego data using CDASC, ICSC, LSDM-MoG, PTA, and LRR.

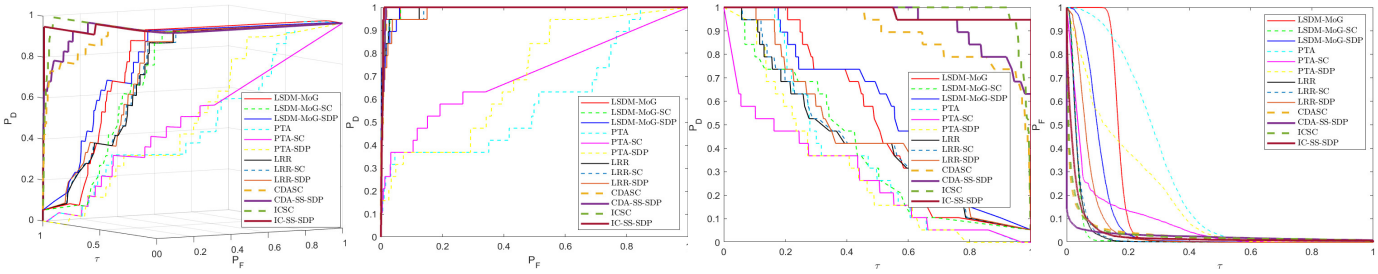


Fig. 22. 3-D ROC curves along with corresponding 2-D ROC curves by using CDASC, ICSC, LSDM-MoG, PTA, and LRR for HYIDCE panel data in Fig. 19.

Figs. 19–21 show the detection results obtained by CDA, IC, LSDM-MoG-MD and PTA using SC and SS-SDP for HYDICE 15-panel scene, HYDICE urban, and San Diego airport datasets, respectively, where the best results of CADSC, CDA-SS-SDP, ICSC, and IC-SS-SDP obtained in Section VII-B are also included for comparison. By visual inspection, it is very clear that LSDM-MoG performed better than PTA for HYDICE panel and urban datasets, while PTA detected three airplanes more visibly than LSDM-MoG but with many falsely alarmed data samples. Nevertheless, on the whole, CADSC/SDP-AD and ICSC/SDP-AD did outperform both LSDM-MoG and PTA, particularly, significantly better than PTA.

To further conduct quantitative analysis, Figs. 22–24 plot the 3-D ROC curves along with their corresponding 2-D ROC curves calculated from the detection maps from Figs. 19 to 21

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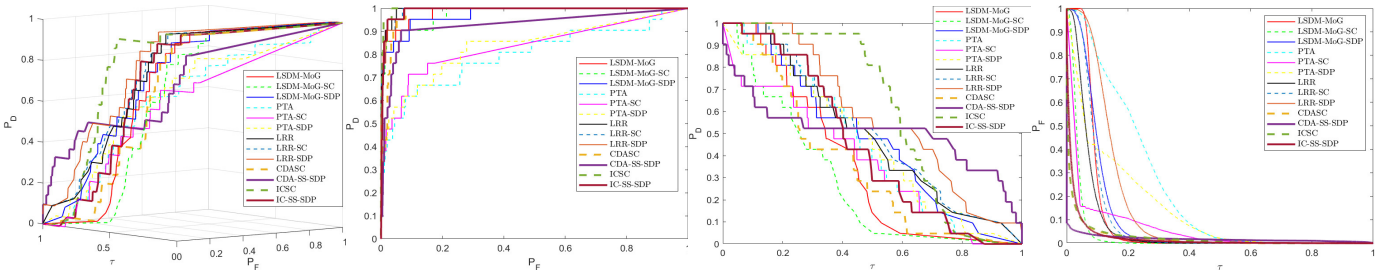


Fig. 23. 3-D ROC curves along with corresponding 2-D ROC curves by CDASC, ICSC, LSDM-MoG, PTA, and LRR for HYDICE URBAN data in Fig. 20.

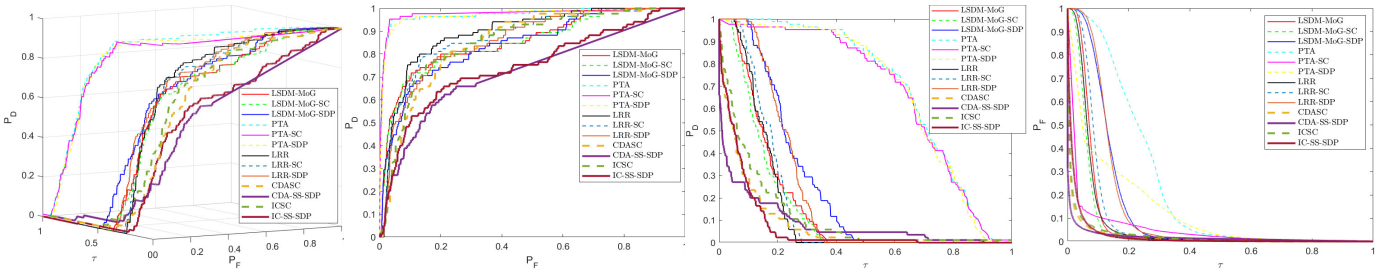


Fig. 24. 3-D ROC curves along with corresponding 2-D ROC curves by CDASC, ICSC, LSDM-MoG, PTA, and LRR for San Diego data in Fig. 21.

TABLE XI

AUC OF SAN DIEGO AIRPORT SCENE PRODUCED IN FIG. 24 BY CDASC, ICSC, LSDM-MOG, PTA, AND LRR WITH $p = 11$ AND $j = 9$

detector	$AUC_{(D,F)}$	$AUC_{(D,\tau)}$	$AUC_{(F,\tau)}$	AUC_{TD}	AUC_{BS}	AUC_{SNPR}	AUC_{TDBS}	AUC_{ODP}	Time
$\delta^{\text{RX-AD}}_{\text{CDASC}}$	**0.8492**	**0.0863**	0.0195	**0.9355**	**0.8296**	4.4150	**0.0668**	**0.9160**	**0.357907 s**
$\delta^{\text{RX-AD}}_{\text{CDA-SS-SDP}}$	0.7225	0.0836	**0.0181**	0.8061	0.7043	**4.6228**	0.0655	0.7880	0.549688 s
$\delta^{\text{RX-AD}}_{\text{ICSC}}$	**0.8530**	**0.1158**	0.0242	**0.9687**	**0.8270**	**4.7894**	**0.0916**	**0.9446**	**0.387074 s**
$\delta^{\text{OSP-AD}}_{\text{IC-SS-SDP}}$	0.7096	0.0106	**0.0028**	0.7203	0.7067	3.8081	0.0078	0.7175	0.492797 s
$\delta^{\text{LSDM-MoG-MD}}$	**0.8643**	0.2035	0.1226	**1.0678**	0.7417	1.6600	0.0809	**0.9452**	0.821390 s
$\delta^{\text{LSDM-MoG-MD}}_{\text{SC}}$	0.8316	0.1373	**0.0654**	0.9689	**0.7661**	**2.0986**	0.0719	0.9035	0.975057 s
$\delta^{\text{LSDM-MoG-MD}}_{\text{SS-SDP}}$	0.8246	**0.2213**	0.1298	1.0459	0.6948	1.7055	**0.0915**	0.9162	1.189272 s
δ^{PTA}	0.9737	**0.6928**	0.2358	**1.6666**	0.7376	2.9379	0.4570	1.4307	0.081479 s
$\delta^{\text{PTA}}_{\text{SC}}$	**0.9765**	0.6741	**0.0523**	1.6506	**0.9239**	**12.8812**	**0.6217**	**1.5982**	0.152978 s
$\delta^{\text{PTA}}_{\text{SS-SDP}}$	0.9714	0.6795	0.1282	1.6510	0.8431	5.2988	0.5513	1.5227	0.352450 s
δ^{LRR}	**0.8770**	0.1418	**0.0724**	1.0188	**0.8046**	**1.9590**	0.0694	**0.9464**	0.104659 s
$\delta^{\text{LRR}}_{\text{SC}}$	0.8401	0.1568	0.0952	0.9969	0.7448	1.6471	0.0616	0.9017	0.182466 s
$\delta^{\text{LRR}}_{\text{SS-SDP}}$	0.8416	**0.2620**	0.1795	**1.1036**	0.6620	1.4596	**0.0825**	0.9241	0.403355 s

where Tables IX–XI tabulate the eight detection measures derived from the 3-D ROC-curves with the best results for each method boldfaced. As we can see, SC and SDP significantly improved the performance of LSDM-MoG, PTA, and LRR, specifically, PTA which performed much worse than LSDM-MoG and LRR.

XII. CONCLUSION

This article presents an approach that combines DS and SDP for AD. It is quite different from current existing AD approaches in the sense that it does not require commonly used LRaSR-based models or LRaSMD-based models to separate BKG and anomalies into low rank and sparse spaces. Two key ideas are contributed to this article. One is to use DS to remove 2OS-characterize BKG and then FastICA to generate a j-IC component space from the sphered data space, $\mathbf{X}_{\text{DS}}$, as a sparse space, $\mathbf{X}_{\text{IC}^j}$, referred to as DS-IC approach which distinguishes itself from the CDA approach developed in [44] using the PC-IC model. The other key idea is to develop a new concept, SDP, to separate anomalies from non-Gaussian noise in $\mathbf{X}_{\text{IC}^j}$ which gives rise to a DS-IC-SDP space. Therefore, one major contribution of this article is to produce a j-IC component space $\mathbf{X}_{\text{IC}^j}$ without requiring any LRaSR-based model. A second major contribution is introduction of SDP which expresses data sample values in either $\mathbf{X}_{\text{DS}}$ or $\mathbf{X}_{\text{IC}^j}$ as a probability vector to represent an SDP-produced

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sparsity density curve. The prominent peaks extracted from the sparsity curve form an SDP-constrained space, called $\mathbf{X}_{\mathrm{DS}}^{\mathrm{SDP}}/\mathbf{X}_{\mathrm{IC}^j}^{\mathrm{SDP}}$, to further improve anomaly detectability. In order to determine an appropriate number of SDP peaks, VD is used in conjunction with MX-SVD for this purpose. Once DS-SDP space and IC-SDP space are found, any AD can be readily applied to these spaces for AD. Interestingly, many current AD methods such as LRR, LSDM-MoG, and PTA can be further improved by DS-SDP/IC-SDP as shown in Section VIII.

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